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# Quantile regression for dynamic panel data with fixed effects

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ABSTRACT

This paper studies a quantile regression dynamic panel model with fixed effects. Panel data fixed effects estimators are typically biased in the presence of lagged dependent variables as regressors. To reduce the dynamic bias, we suggest the use of the instrumental variables quantile regression method of Chernozhukov and Hansen (2006) along with lagged regressors as instruments. In addition, we describe how to employ the estimated models for prediction. Monte Carlo simulations show evidence that the instrumental variables approach sharply reduces the dynamic bias, and the empirical levels for prediction intervals are very close to nominal levels. Finally, we illustrate the procedures with an application to forecasting output growth rates for 18 OECD countries.



## 1. Introduction

Recently, there has been a growing literature on estimation, testing, and prediction using semiparametric panel data models. Koenker (2004) introduced a general approach to estimation of quantile regression (QR) models for longitudinal data. Individual specific (fixed) effects are treated as pure location shift parameters common to all conditional quantiles and may be subject to shrinkage toward a common value as in the Gaussian random effects paradigm. Controlling for individual specific heterogeneity via fixed effects (FE) while exploring heterogeneous covariate effects within the QR framework offers a more flexible approach to the analysis of panel data than that afforded by the classical Gaussian fixed and random effects estimators. Recent work by Lamarche (2010) and Geraci and Bottai (2007) has elaborated on this form of penalized QR estimator. Abrevaya and Dahl (2008) have introduced an alternative quantile-estimation approach motivated by a correlated random-effects model.

In econometric applications the modeling of dynamic relationships and the availability of panel data often suggest dynamic model specifications involving lagged dependent variables. Consistency of estimators in conventional dynamic panel data models depends critically on the assumptions about the initial conditions of the dynamic process. It has been recognized at least since Nickell (1981) that classical ordinary least squares (OLS) estimators in dynamic panel models with FE are seriously biased when the temporal dimension of the panel is short. An extensive literature, initiated by Anderson and Hsiao (1981, 1982) and Arellano and Bond (1991), has explored instrumental variables (IV) approaches to attenuate the bias, showing that IV methods are able to produce consistent estimators that are independent of the initial conditions.[1]

Conventional QR estimation of dynamic panel data models with fixed effects suffers from similar bias effects as those seen in the least squares case when the time dimension is modest. Reliance on the existing least squares strategies for bias reduction is unsatisfactory in the QR setting for at least two reasons. First, differencing is inappropriate, either temporally or via the usual deviation from individual means (within) transformation. Linear transformations that are completely innocuous in the context of conditional mean models are highly problematic in the conditional quantile models since they alter in a fundamental way what is being estimated.[2] Second, the implementation of the IV method needs to be rethought. Fortunately, neither problem is insurmountable. There is no need to transform the QR model to compute the FE estimator. This is a computable convenience in the least squares case, but even when the number of FE is large, interior

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[1] See, e.g., Ahn and Schmidt (1995), Blundell and Bond (1998) and Bun and Carree (2005). More recently, a number of additional approaches have been proposed to reduce the bias in dynamic and nonlinear panels. These methods use asymptotic approximations derived as both the number of individuals, $N$, and the number time series, $T$, go to infinity jointly; see e.g. Arellano and Hahn (2007) for a survey, and Hahn and Kuersteiner (2002), Alvarez and Arellano (2003), Hahn and Newey (2004) and Bester and Hansen (2009), for specific approaches.

[2] Expectations enjoy the convenient property that they commute with linear transformations; quantiles do not. This intrinsic difficulty has been recognized by Abrevaya and Dahl (2008), among others, and is clarified by Koenker and Hallock (2000, p. 19): "Quantiles of convolutions of random variables are rather intractable objects, and preliminary differencing strategies familiar from Gaussian models have sometimes unanticipated effects".

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point optimization methods using modern sparse linear algebra make direct estimation of the QR model quite efficient.

This paper investigates estimation, inference, and prediction in a quantile regression formulation of the dynamic panel data model with individual specific intercepts. We find that conventional FE estimation of the QR specification suffers from similar bias problems as those of the least squares estimation. To reduce the dynamic bias in the quantile regression FE estimator, we suggest the use of the IV quantile regression method of Chernozhukov and Hansen (2005, 2006, 2008) along with lagged (or lagged differences of the) regressors as instruments. Thus, the estimator combines the usual IV concept for dynamic panel data and the quantile regression IV framework.

There is an emerging literature on forecasting with dynamic panel data (see e.g. Baltagi (2008)), and an important application of the proposed panel model is out-of-sample prediction. Most econometric forecasting methods using panels have focused on models for the conditional mean under Gaussian conditions. The quantile regression model has a significant advantage over these models, since it will be less sensitive to the tail behavior of the underlying random variables representing the forecasting variable of interest, and consequently will be less sensitive to outliers. Moreover, because of the heterogeneous nature of most of the variables of interest in economics, prediction using QR techniques is an important tool for applied work. The model proposed in this paper can also be used to predict the conditional density function of the variable of interest under weak assumptions.

Monte Carlo studies are conducted to evaluate the finite sample properties of the estimator. The simulations show that the quantile regression FE estimator is significantly biased in the presence of lagged dependent variables, while the IV method sharply reduces the bias. The experiments suggest that the quantile regression IV approach for dynamic panel data turns out to be especially advantageous when innovations are heavy-tailed. In addition, we conduct simulations to evaluate the performance of the prediction intervals. The results provide evidence that the empirical levels approximate well the nominal or theoretical levels, and the length of the prediction interval decreases with the sample size.

Finally, we illustrate the methods with an application to forecasting output growth rates for 18 OECD countries during the time period that includes the subprime economic crisis. We compute one-step-ahead prediction intervals and density functions. The results from estimation of an autoregressive dynamic model show that for most countries zero growth rate lies inside the interval with 0.90 nominal level. In a second round of estimation, we also include the stock price index as an exogenous covariate. In this case, the results show negative lower bound for all countries, and the densities forecast have more mass on the left tail.

The rest of the paper is organized as follows. Section 2 presents the quantile regression dynamic panel data IV with FE estimation and inference. In Section 3 we describe how to employ the estimated models for prediction. Section 4 describes the Monte Carlo experiments. In Section 5, we briefly illustrate the new approach. Finally, Section 6 concludes the paper.

## 2. The model and assumptions

### 2.1. Estimation

In this section we introduce estimation of the quantile regression dynamic panel data instrumental variables (QRPIV) that includes individual specific FE. Consider the classical dynamic model for panel data with individual fixed effects[3]

$$y_{it} = \eta_i + \alpha y_{it-1} + x_{it}'\beta + u_{it} \quad i = 1, \dots, N;\ t = 1, \dots, T \qquad (1)$$

where $y_{it}$ is the response variable, $\eta_i$ denotes the individual FE, $y_{it-1}$ is the lag of the response variable, $x_{it}$ is a $p$-vector of exogenous covariates, and $u_{it}$ is the innovation term. It is possible to write model (1) in a more concise matrix form as,

$$y = Z\eta + \alpha y_{-1} + X\beta + u, \qquad (2)$$

so that $Z = I_N \otimes \iota_T$, $\iota_T$ is a $T \times 1$ vector of ones, and $\eta = (\eta_1, \dots, \eta_N)'$ is the $N \times 1$ vector of individual specific effects or intercepts. Note that $Z$ represents an incidence matrix that identifies the $N$ distinct individuals in the sample.

The analogous version model to (1) for the $\tau$th conditional quantile function of the response of the $t$th observation on the $i$th individual $y_{it}$ can be represented as

$$Q_{y_{it}}(\tau | y_{it-1}, x_{it}) = \eta_i + \alpha(\tau) y_{it-1} + x_{it}'\beta(\tau). \qquad (3)$$

In model (3) only the effects of the covariates $(y_{it-1}, x_{it})$ are allowed to depend upon the quantile, $\tau$, of interest. The $\eta$'s are intended to capture some individual specific source of variability, or "unobserved heterogeneity," that was not adequately controlled for other covariates. In most applications the time series dimension $T$ is relatively small compared to the number of individuals $N$. Therefore, it might be difficult to estimate a $\tau$-dependent distributional individual effect, and we restrict the estimates of the individual specific effects to be independent of $\tau$ across the quantiles. The restriction can be implemented by estimating the model for several quantiles simultaneously.

Koenker (2004) introduced a general approach to estimation of quantile regression FE models for panel data. Applying this principle to Eq. (3), one would solve

$$(\hat{\eta}, \hat{\alpha}, \hat{\beta}) = \min_{\eta,\alpha,\beta} \sum_{k=1}^{K} \sum_{i=1}^{N} \sum_{t=1}^{T} \upsilon_k \rho_\tau \times (y_{it} - \eta_i - \alpha(\tau_k) y_{it-1} - x_{it}'\beta(\tau_k))$$

where $\rho_\tau(u) := u(\tau - I(u < 0))$ as in Koenker and Bassett (1978), and $\upsilon_k$ are the weights that control the relative influence of the $K$ quantiles $\{\tau_1, \dots, \tau_K\}$ on the estimation of the $\eta_i$.

However, we will see that the quantile regression FE estimator, as in the OLS case, is biased in the presence of lagged dependent variables as regressors. In least squares estimation of dynamic panel models it is evident that the unobserved initial values of the dynamic process induce a bias.[4] For long panels, the effect associated with the initial conditions is seen to be $O(T^{-1})$ and therefore negligible. Latter we evaluate the dynamic bias in the within-group and quantile regression FE estimators by means of Monte Carlo simulation. We find that the quantile regression FE estimator suffers from similar bias effects to those seen in the least squares case when $T$ is moderate.

Anderson and Hsiao (1981, 1982) and Arellano and Bond (1991), in the linear regression case, show that IV methods are able to produce consistent estimators for dynamic panel data models that are independent of the initial conditions. These estimators are based on the idea that lagged (or lagged differences of) the regressors are correlated with the included regressor but are uncorrelated with the innovations. Thus, valid instruments, $w_{it}$, are available from inside the model and can be used to estimate the parameters of interest by IV methods. In this paper we use an analogous rationality for the construction of instruments.

The problem of bias for the dynamic panel quantile regression can be ameliorated through the use of instrumental variables, $w$, that affect the determination of lagged $y$ but are independent of

[3] To simplify the presentation we focus on the first-order autoregressive processes, since the main insights generalize in a simple way to higher-order cases.

[4] See Hsiao (2003), and Arellano (2003) for more details. Nickell (1981) provides analytical calculations for bias in the within-group estimator in a linear dynamic panel model.

innovations. Following Chernozhukov and Hansen (2006, 2008), and assuming the availability of instrumental variables, $w_{it}$, we consider estimators defined as

$$\hat{\alpha} = \min_{\alpha} \|\hat{\gamma}(\alpha)\|_A, \tag{4}$$

where

$$(\hat{\eta}(\alpha), \hat{\beta}(\alpha), \hat{\gamma}(\alpha)) = \min_{\eta,\beta,\gamma} \sum_{k=1}^{K}\sum_{i=1}^{N}\sum_{t=1}^{T} \upsilon_k \rho_\tau (y_{it} - \eta_i - \alpha(\tau_k) y_{it-1} - x'_{it}\beta(\tau_k) - w'_{it}\gamma(\tau_k)), \tag{5}$$

with $\|x\|_A = \sqrt{x'Ax}$, and $A$ is a positive definite matrix. Our final parameter estimates of interest are thus

$$\hat{\theta}(\tau) \equiv (\hat{\alpha}(\tau), \hat{\beta}(\tau)) \equiv (\hat{\alpha}(\tau), \hat{\beta}(\hat{\alpha}(\tau), \tau)). \tag{6}$$

The intuition underlying the estimator is that, since $w$ is a valid instrument and it is independent of $u$, it should have a zero coefficient. The estimator (6) finds parameter values for $\alpha$ and $\beta$ through the inverse step (4) such that the value of coefficient $\gamma(\alpha, \tau)$ on $w$ in the ordinary QR step (5) is driven as close to zero as possible. Hence, by minimizing the coefficient of the variable $w_{it}$ one can recover the estimator of $\alpha$. Therefore, the bias generated by inclusion of $y_{it-1}$ in Eq. (3) is reduced through the presence of IV, $w_{it}$, that affect the determination of $y_{it}$ but are independent of $u_{it}$. Values of $y$ lagged (or differences) two periods or more and/or lags of the exogenous variable $x$ affect the determination of lagged $y$ but are independent of $u$, so they can potentially be used as instruments to estimate $\alpha$ and $\beta$ by the QRPIV method. As suggested by Chernozhukov and Hansen (2008), in practice, a simple procedure is to let the instruments $w_{it}$ either be $w_{it}$ or the predicted value from a least squares projection of lagged $y$ on $w$ and $x$.

The implementation of the QRPIV procedure is straightforward. Define the objective function

$$R_{NT}(\tau, \eta_i, \alpha, \beta, \gamma) := \sum_{k=1}^{K}\sum_{i=1}^{N}\sum_{t=1}^{T} \upsilon_k \rho_\tau (y_{it} - \eta_i - \alpha(\tau_k) y_{it-1} - x'_{it}\beta(\tau_k) - w'_{it}\gamma(\tau_k))$$

where $y_{it-1}$ is, in general, a $dim(\alpha)$-vector of endogenous variables, $\eta_i$ are the FE, $x_{it}$ is a $dim(\beta)$-vector of exogenous explanatory variables, $w_{it}$ is a $dim(\gamma)$-vector of IV such that $dim(\gamma) \geq dim(\alpha)$. For the special case of $K = 1$, one can use a grid search as follows:

(1) For a given quantile of interest $\tau$, define a grid of values $\{\alpha_j, j = 1, \ldots, J; |\alpha| < 1\}$, and run the ordinary $\tau$-quantile regression of $(y_{it} - \alpha_j y_{it-1})$ on $(z_{it}, w_{it}, x_{it})$ to obtain coefficients $\hat{\eta}(\alpha_j, \tau)$, $\hat{\beta}(\alpha_j, \tau)$ and $\hat{\gamma}(\alpha_j, \tau)$; that is, for a given value of the autoregression structural parameter, say $\alpha$, one estimates the ordinary panel QR to obtain

$$(\hat{\eta}_i(\alpha_j, \tau), \hat{\beta}(\alpha_j, \tau), \hat{\gamma}(\alpha_j, \tau)) := \min_{\eta_i,\beta,\gamma} R_{NT}(\tau, \eta_i, \alpha, \beta, \gamma).$$

(2) To find an estimate for $\alpha(\tau)$, choose $\hat{\alpha}(\tau)$ as the value among $\{\alpha_j, j = 1, \ldots, J\}$ that makes $\|\hat{\gamma}(\alpha_j, \tau)\|$ closest to zero. Formally, let

$$\hat{\alpha}(\tau) = \min_{\alpha \in A} [\hat{\gamma}(\alpha, \tau)']\hat{A}(\tau)[\hat{\gamma}(\alpha, \tau)]$$

where $A$ is a positive definite matrix. The estimate $\hat{\beta}(\tau)$ is then given by $\hat{\beta}(\hat{\alpha}(\tau), \tau)$, which leads to the estimates: $\hat{\theta}(\tau) = (\hat{\alpha}(\tau), \hat{\beta}(\tau)) = (\hat{\alpha}(\tau), \hat{\beta}(\hat{\alpha}(\tau), \tau))$.

For the case of $K > 1$, the optimization is very large depending on the number of estimated quantiles. Therefore, instead of using a grid search, we use a numerical optimization function in R. As starting values, we use the estimates from the QR FE model without any instruments. The design matrix is $[\upsilon \otimes (I_N \otimes \iota_T) \vdots \Upsilon \otimes y_{-1} \vdots \Upsilon \otimes X \vdots \Upsilon \otimes w]$, where $I_N$ is a $N \times N$ identity matrix, $\iota_T$ is a $T \times 1$ vector of ones, $\Upsilon$ is a $K \times K$ diagonal matrix with the weights $\upsilon$. The response vector is $\tilde{y} = (\upsilon \otimes y)$. As Koenker (2004) observes, in typical applications the design matrix of the full problem is very sparse, i.e. has mostly zero elements, and this considerably reduces the computational effort and memory requirements in large problems.

The existence of the parameter $\eta_i$, whose dimension $N$ tends to infinity, raises some new issues for the asymptotic analysis of the proposed estimator. As first noted by Neyman and Scott (1948), leaving the individual heterogeneity unrestricted in a nonlinear or dynamic model generally results in inconsistent estimators of the common parameters due to the incidental parameters problem; that is, noise in the estimation of the fixed effects when the time dimension is short results in inconsistent estimates of the common parameters due to the nonlinearity of the problem. In this respect, QR panel data suffers from this problem. Koenker (2004) overcomes this difficulty by using a large $N$ and $T$ asymptotics. We impose the following regularity conditions:

**A1.** The $y_{it}$ are independent across individuals, covariance stationary, with conditional distribution functions $F_{it}$, and differentiable conditional densities, $0 < f_{it} < \infty$, with bounded derivatives $f'_{it}$ for $i = 1, \ldots, N$ and $t = 1, \ldots, T$.

**A2.** Let $y = (y_{it})$, $Z = I_N \otimes \iota_T$, and $\iota_T$ a $T$-vector of ones, $y_{-1} = (y_{it-1})$ be a $NT \times dim(\alpha)$ matrix, $X = (x_{it})$ be a $NT \times dim(\beta)$ matrix, and $W = (w_{it})$ be a $NT \times dim(\gamma)$ matrix. For

$$\Pi(\eta, \alpha, \beta, \tau) := E[\upsilon(\tau - 1(y < Z\eta + y_{-1}\alpha + X\beta))\check{X}(\tau)],$$

$$\Pi(\eta, \alpha, \beta, \gamma, \tau) := E[\upsilon(\tau - 1(y < Z\eta + y_{-1}\alpha + X\beta + W\gamma))\check{X}(\tau)],$$

$\check{X}(\tau) := [Z, W, X]'$, the Jacobian matrices $\frac{\partial}{\partial(\eta,\alpha,\beta)}\Pi(\eta, \alpha, \beta, \tau)$ and $\frac{\partial}{\partial(\eta,\beta,\gamma)}\Pi(\eta, \alpha, \beta, \gamma, \tau)$ are continuous and have full rank uniformly over $\mathscr{E} \times \mathscr{A} \times \mathscr{B} \times \mathscr{G} \times \mathscr{T}$. The parameter space, $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$, is a connected set. Moreover the image of $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$ under the map $(\eta, \alpha, \beta) \to \Pi(\eta, \alpha, \beta, \tau)$ is simply connected;

**A3.** Denote $\Phi(\tau_k) = diag(f_{it}(\xi_{it}(\tau_k)))$, where $\xi_{it}(\tau_k) = \eta_i + \alpha(\tau_k)y_{it-1} + x'_{it}\beta(\tau_k) + w'_{it}\gamma(\tau_k)$, $M_{Z_k} = I - P_{Z_k}$ and $P_{Z_k} = Z(Z'\Phi(\tau_k)Z)^{-1}Z'\Phi(\tau_k)$. Let $\tilde{X} = [W', X']'$. Then, the following matrices are positive definite:

$$J_\vartheta = \lim_{N,T\to\infty} \frac{1}{NT} \times \begin{pmatrix} \upsilon_1 \tilde{X}'M'_{Z_1}\Phi(\tau_1)M_{Z_1}\tilde{X} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \upsilon_k \tilde{X}'M'_{Z_k}\Phi(\tau_k)M_{Z_k}\tilde{X} \end{pmatrix}$$

$$J_\alpha = \lim_{N,T\to\infty} \frac{1}{NT} \times \begin{pmatrix} \upsilon_1 \tilde{X}'M'_{Z_1}\Phi(\tau_1)M_{Z_1}y_{-1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \upsilon_k \tilde{X}'M'_{Z_k}\Phi(\tau_k)M_{Z_k}y_{-1} \end{pmatrix},$$

and

$$S = \lim_{N,T\to\infty} \frac{1}{NT} \begin{pmatrix} \Sigma_{11}\tilde{X}'M'_{Z_1}M_{Z_1}\tilde{X} & \cdots & \Sigma_{1k}\tilde{X}'M'_{Z_1}M_{Z_k}\tilde{X} \\ \vdots & \ddots & \vdots \\ \Sigma_{k1}\tilde{X}'M'_{Z_k}M_{Z_1}\tilde{X} & \cdots & \Sigma_{kk}\tilde{X}'M'_{Z_k}M_{Z_k}\tilde{X} \end{pmatrix},$$

where $\Sigma_{ij} = \upsilon_i(\tau_i \wedge \tau_j - \tau_i\tau_j)\upsilon_j$. Now define $[\bar{J}'_\beta, \bar{J}'_\gamma]'$ as a partition of $J_\vartheta^{-1}$, and $H = \bar{J}'_\gamma A[\alpha(\tau_k)]\bar{J}_\gamma$. Then, $J_\vartheta$ is invertible, and $J'_\alpha H J_\alpha$ is also invertible;

**A4.** For all $\tau \in \mathcal{T} = [c, 1-c]$ with $c \in (0, 1/2)$, $(\alpha(\tau), \beta(\tau)) \in \text{int}\, \mathcal{A} \times \mathcal{B}$, and $\mathcal{A} \times \mathcal{B}$ is compact and convex;

**A5.** $\max_{it} \|y_{it}\| = O(\sqrt{NT})$; $\max_{it} \|x_{it}\| = O(\sqrt{NT})$; $\max_{it} \|w_{it}\| = O(\sqrt{NT})$;

**A6.** $T \to \infty$ as $N \to \infty$ and $\frac{N^a}{T} \to 0$ for some $a > 0$.

Condition A1 is a standard assumption in the quantile regression literature and imposes a restriction on the density function of $y_{it}$. Condition A2 is important for identification of the parameters. The continuity and full rank conditions require that the instrument $W$ impacts the conditional distribution of $y$ at many relevant points. Assumption A3 states conditions for the matrices that guarantee asymptotic normality. A4 imposes compactness on the parameter space of $\alpha(\tau)$. Such an assumption is needed since the objective function is not convex in $\alpha$. Assumption A5 imposes bounds on the variables. Finally, condition A6 is the same assumption as in Koenker (2004) and Lamarche (2010).

To further comment on the nature of the correlation between $y_{-1}$ and $W$ required by A2, note that, for a given quantile $\tau$, by A1 we have that

$$\partial E[(\tau - 1(y < Z\eta + y_{-1}\alpha + X\beta))\check{X}(\tau)]/\partial(\eta, \alpha, \beta)$$
$$= E[(Z', W', X')'\Phi(Z', y_{-1}', X')].$$

Hence, the Jacobian in A2 takes a form of density-weighted covariance matrix for $Z, y_{-1}$ and $W$ variables, and requires that this matrix has full rank. In addition, A2 imposes that global identifiability must hold; hence, the impact of $W$ should be rich enough to guarantee that the equations are solved uniquely.

We can now establish consistency and asymptotic normality of the estimator. Proofs appear in the Appendix.

**Theorem 1.** *Given assumptions* A1–A6, $(\eta, \alpha(\tau), \beta(\tau))$ *uniquely solves the equations* $E[\upsilon\psi(y - Z\eta - y_{-1}\alpha - X\beta)\check{X}(\tau)] = 0$ *over* $\mathcal{E} \times \mathcal{A} \times \mathcal{B}$, *and* $\theta(\tau) = (\alpha(\tau), \beta(\tau))$ *is consistently estimable.*

The limiting distribution of the parameters of interest is given in Theorem 2.

**Theorem 2.** *Under conditions* A1–A6, *for a given* $\tau \in (0, 1)$, $\hat{\theta}$ *converges to a Gaussian distribution:*

$$\sqrt{NT}(\hat{\theta}(\tau) - \theta(\tau)) \xrightarrow{d} N(0, \Omega(\tau)),$$
$$\Omega(\tau) = (K', L')'S(K', L')$$

*where* $S = (\min(\tau, \tau') - \tau\tau')E(VV')$, $V = (\upsilon_1\tilde{X}'M_{Z_1}', \ldots, \upsilon_k\tilde{X}'M_{Z_k}')'$, $K = (J_\alpha' H J_\alpha)^{-1} J_\alpha H$, $H = \bar{J}_\gamma' A[\alpha(\tau)]\bar{J}_\gamma$, $L = \bar{J}_\beta M$, $M = I - J_\alpha K$, $[\bar{J}_\beta, \bar{J}_\gamma]$ *is a partition of* $J_\vartheta$, $\Phi(\tau_k) = diag(f_{it}(\xi_{it}(\tau_k)))$, $\tilde{X} = [W', X']'$, *and* $J_\vartheta$ *and* $J_\alpha$ *are as defined in assumption* A3.

**Remark 1.** When $dim(\gamma) = dim(\alpha)$, the choice of $A(\alpha)$ does not affect the asymptotic variance. As in Chernozhukov and Hansen (2008), when $dim(\gamma) > dim(\alpha)$, the choice of the weighting matrix $A(\alpha)$ generally matters, and it is important for efficiency. A natural choice for $A(\alpha)$ is given by the inverse of the covariance matrix of $\hat{\gamma}(\alpha(\tau), \tau)$. Noticing that $A(\alpha)$ is equal to $(\bar{J}_\gamma S \bar{J}_\gamma)^{-1}$ at $\alpha(\tau)$, it follows that the asymptotic variance of $\sqrt{NT}(\hat{\alpha}(\tau) - \alpha(\tau))$ is given by $\Omega_\alpha = (J_\alpha' \bar{J}_\gamma' (\bar{J}_\gamma S \bar{J}_\gamma')^{-1} \bar{J}_\gamma J_\alpha)^{-1}$.

The components of the asymptotic variance matrix that need to be estimated include $J_\vartheta$, $J_\alpha$ and $S$. The matrix $S$ can be estimated by its sample counterpart

$$\hat{S}(\tau, \tau') = (\min(\tau, \tau') - \tau\tau')\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} V_{it}V_{it}'.$$

Following Powell (1986), $J_\vartheta$ and $J_\alpha$ can be estimated as stated in Theorem 2. The typical element of $\hat{J}_\vartheta$ takes the following form

$$\hat{J}_{\vartheta_j} = \upsilon_j \frac{1}{2NTh_n}\sum_{i=1}^{N}\sum_{t=1}^{T} I(|\hat{u}(\tau_j)| \le h_n)\tilde{X}M_{Z_j}M_{Z_j}'\tilde{X}'$$

where $\hat{u}(\tau_j) \equiv y - Z\hat{\eta} - \hat{\alpha}(\tau_j)y_{-1} - X\hat{\beta}(\tau_j)$ and $h_n$ is an appropriately chosen bandwidth. The estimator of $\hat{J}_{\alpha_j}$ is analogous to $\hat{J}_{\vartheta_j}$. Using the same procedure we can estimate the element $Z\hat{\Phi}(\tau_j)Z$ in $P_Z$. The consistency of these asymptotic covariance matrix estimators will not be discussed further in this paper.

### 2.2. Inference

Now we turn our attention to inference in the quantile regression dynamic panel instrumental variable (QRPIV) model, and suggest Wald and Kolmogorov–Smirnov type tests for general linear hypotheses. In the independent and identically distributed setting the conditional quantile functions of the response variable, given the covariates, are all parallel, implying that covariate effects shift the location of the response distribution but do not change the scale or shape. However, slope estimates often vary across quantiles implying that it is important to test for equality of slopes across quantiles. Wald tests designed for this purpose were suggested by Koenker and Bassett (1982), and Koenker and Machado (1999). It is possible to formulate a wide variety of tests using variants of the proposed Wald test, from simple tests on a single QR coefficient to joint tests involving many covariates and distinct quantiles at the same time. The Wald process and associated limiting theory provide a natural foundation for the hypothesis $R\theta(\tau) = r$ when $r$ is known.

Gutenbrunner and Jureckova (1992) show that the QR process is tight and thus the limiting variate viewed as a function of $\tau$ is a Brownian Bridge over $\tau \in \mathcal{T}$.[5] Therefore, under the linear hypothesis $H_0 : R\theta(\tau) = r$, and letting $\Gamma = (K', L')'EVV'(K', L')$,

$$\mathcal{V}_{NT} = \sqrt{NT}[R\Gamma(\tau)R']^{-1/2}(R\hat{\theta}(\tau) - r) \Rightarrow B_q(\tau), \tag{7}$$

where $B_q(\tau)$ represents a $q$-dimensional standard Brownian Bridge. For any fixed $\tau$, $B_q(\tau)$ is $N(0, \tau(1-\tau)I_q)$. The normalized Euclidean norm of $B_q(\tau)$, $Q_q(\tau) = \|B_q(\tau)\|/\sqrt{\tau(1-\tau)}$, is generally referred to as a Kiefer process of order $q$. Thus, for given $\tau$, the regression Wald process can be constructed as

$$\mathcal{W}_{NT} = NT(R\hat{\theta}(\tau) - r)'[R\hat{\Omega}(\tau)R']^{-1}(R\hat{\theta}(\tau) - r), \tag{8}$$

where $\hat{\Omega}$ is a consistent estimator of $\Omega$, which is given in Theorem 2.

If we are interested in testing $R\theta(\tau) = r$ at a particular quantile $\tau = \tau_0$, a Chi-square test can be conducted based on the statistic $\mathcal{W}_{NT}(\tau_0)$. Under $H_0$, given a consistent estimate of $\hat{\Omega}(\tau)$, the statistic $\mathcal{W}_{NT}$ is asymptotically $\chi_q^2$ with $q$-degrees of freedom, where $q$ is the rank of the matrix $R$. The limiting distribution of the test is summarized as follows.

**Theorem 3** (*Wald Test*)**.** *Under* $H_0 : R\theta(\tau) = r$, *and conditions* A1–A6, *for fixed* $\tau$,

$$\mathcal{W}_{NT}(\tau) \overset{a}{\sim} \chi_q^2.$$

**Proof.** The proof of Theorem 3 follows directly from Eq. (8) and Theorem 2. □

It is important to note that given a consistent estimate of the variance-covariance matrix, one can easily construct confidence

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5 In a related result, Wei and He (2006) establish tightness of the QR process in the longitudinal data context with increasing parameter dimension, see for instance, Lemma 8.4.

interval estimates for the parameters of interest based on the asymptotic approximation given in Theorem 2, or inverting the above Wald test.

More general hypotheses are also easily accommodated by the Wald approach. Let $\zeta = (\theta(\tau_1)', \ldots, \theta(\tau_m)')$ and define the null hypothesis as $H_0 : R\nu = r$. The test statistic is the same Wald test as Eq. (8). In this case $\Omega$ is the matrix with $(k, l)$th block

$$\Omega(\tau_k, \tau_l) = (K'(\tau_k), L'(\tau_k))' S(\tau_k, \tau_l)(K'(\tau_l), L'(\tau_l)),$$

where $\Omega(\tau_k, \tau_l)$ is as defined in Theorem 2. The statistic $\mathcal{W}_{NT}$ is still asymptotically $\chi_q^2$ under $H_0$ where $q$ is the rank of the matrix $R$. This formulation accommodates a wide variety of testing situations, from a simple test on single QR coefficients to joint tests involving several covariates and distinct quantiles.

Another important class of tests involves the Kolmogorov–Smirnov ($KS$) type tests, where the goal is to examine the property of the estimator over a range of quantiles $\tau \in \mathcal{T}$. Thus, if one has interest in testing $R\theta(\tau) = r$ over $\tau \in \mathcal{T}$, one may consider the $KS$ type sup-Wald test. Following Koenker and Xiao (2006), we may construct a $KS$ type test as

$$KS\mathcal{W}_{NT} = \sup_{\tau \in \mathcal{T}} \mathcal{W}_{NT}(\tau). \tag{9}$$

The limiting distribution of the $KS$ test is given by

**Theorem 4** (*Kolmogorov–Smirnov Test*)**.** *Under $H_0$ and conditions* A1–A6,

$$KS\mathcal{W}_{NT} = \sup_{\tau \in \mathcal{T}} \mathcal{W}_{NT}(\tau) \Rightarrow \sup_{\tau \in \mathcal{T}} Q_q^2(\tau).$$

**Proof.** The proof of Theorem 4 follows directly from the continuous mapping theorem and Eqs. (7) and (9). □

Critical values for $\sup Q_q^2(\tau)$ have been tabled by Andrews (1993).

## 3. Prediction

In this section, we propose to use the quantile regression dynamic panel data instrumental variables model for prediction. We first suggest a procedure that uses the quantiles of the empirical distribution of the forecasts to construct prediction intervals. The QR model has a significant advantage over models based on the conditional mean, since it will be less sensitive to the tail behavior of the underlying random variables representing the forecasting variable of interest, and consequently will be less sensitive to observed outliers. Moreover, because of the heterogeneous nature of most of the variables of interest in economics, prediction using QR techniques is an important tool for applied work. Second, we construct conditional density forecasts of the variable of interest under weak assumptions, such conditional density forecasts can be made based on an ensemble of the forecast paths.

The literature on prediction using QR is still growing. Koenker and Bassett (2010) adapt QR methods to the paired comparison framework, the model is capable of delivering estimates of the predictive conditional joint density. Giacomini and Komunjer (2005) construct a conditional quantile forecast encompassing test for evaluation and combination of the conditional quantile forecast. Zhou and Portnoy (1996) suggest direct and studentization methods to construct confidence and prediction intervals for conditional quantile functions. In a companion paper, Zhou and Portnoy (1998) extend these results to a class of heteroscedastic linear models. Koenker and Zhao (1996) show that QR offers a natural approach to the construction of prediction intervals for ARCH-type models. Taylor and Bunn (1999) present an alternative QR approach for construction of predictive distributions and intervals.

There is a substantial amount of literature on forecasting with dynamic panel data. Most of the literature is based on models for conditional mean, with the slope coefficients being homogeneous or individually heterogeneous. Baltagi and Griffin (1997), Hsiao and Tahmiscioglu (1997), Baltagi et al. (2000), and Hoogstrate et al. (2000) present empirical applications, discuss and compare the forecast performance of homogeneous and individually heterogeneous slope estimators.[6] However, as Baltagi (2008) observes, the consistent finding is that the models with homogeneous slope coefficients perform better in forecast performance mostly due to their simplicity, their parsimonious representation and stability of parameters estimates.

An important application of dynamic panel data QR models is out-of-sample prediction. In many situations it is important to construct a prediction interval for a future response variable. The QRPIV model is able to examine how covariates influence the location, scale, and shape of the entire response distribution. Most econometric forecasting methods using panels have focused on models for the conditional mean under Gaussian conditions. The dynamic panel data QR models described above offer an opportunity to significantly expand the scope of forecasting applications. There are several potential applications, for instance, the model might be useful to predict the dynamic demand of electricity, natural-gas or cigarettes across US states. The model can also be used to forecast output growth, investment (based on Tobin's $q$ theory), or exchange rate. The scope of potential examples is substantial.

One-step ahead prediction of the quantile function of $y_{it}$ for an individual $i$ are immediately available from model (3). From a date $T$, when the information is available, using the estimated model, it is possible to generate a forecast for $T+1$ as

$$\hat{Q}_{y_{iT+1}}(\tau | y_{iT}, x_{iT+1}) = \hat{\eta}_i + \hat{\alpha}(\tau) y_{iT} + x'_{iT+1}\hat{\beta}(\tau), \tag{10}$$

where the predictors $(y_{iT}, x_{iT+1})$ are known. Moreover, based on the results of the previous sections, for a fixed quantile $\tau$, the one-step-ahead predictor follows a normal distribution. Let $\check{X} = [Z', W', X']'$, $\delta(\tau) = (\eta, \alpha, \beta)'$, and the standardization matrix $D_{NT} = diag(\sqrt{T}, \sqrt{NT}, \sqrt{NT})$. From a simple extension of the results in Theorem 2, we have that $D_{NT}(\hat{\delta}(\tau) - \delta(\tau)) \xrightarrow{d} N(0, \Sigma(\tau))$, where $\Sigma(\tau) = J(\tau)^{-1} S(\tau) J(\tau)^{-1}$, with $J = E(\check{X}\Phi [Z, y_{-1}, X])$ and $S = \tau(1-\tau)E[\check{X}\check{X}']$. Now, let $\dot{X} = [1, y_T, x_{T+1}]'$, and note that $\hat{Q}_{y_{iT+1}}(\tau) = \dot{X}'\delta(\tau)$, such that $(\hat{Q}_{y_{iT+1}}(\tau) - Q_{y_{iT+1}}(\tau)) = (\dot{X}'\hat{\delta}(\tau) - \dot{X}'\delta(\tau)) \overset{a}{\sim} N(0, \Omega(\tau))$, where $\Omega(\tau) = \dot{X} D_{NT}^{-1} \Sigma(\tau) D_{NT}^{-1} \dot{X}'$. Given this distributional property, we can construct an interval for predictive quantiles.

**Corollary 1.** *Under assumptions of Theorem 2, the $(1-2\lambda)100\%$ interval for the one-step-ahead conditional quantile is of the form*

$$I = (\hat{Q}_{y_{iT+1}}(\tau) - a_{NT}, \hat{Q}_{y_{iT+1}}(\tau) + a_{NT})$$

*where $a_{NT} = z_\lambda \Omega(\tau)^{1/2}$, and $z_\lambda$ denotes the $(1-\lambda)$th standard normal percentile point.*

The proof of the corollary is similar to Theorem 2, and it follows directly from asymptotic normality of the estimator.

In many applications it is important to establish confidence intervals for $s$-step-ahead forecasts. However, the problem of deriving the distribution of this predictor becomes a more difficult task.[7] To solve this problem, Schmidt (1974), in a least squares dynamic model context, proposes to iterate the model and use the

6 See e.g. Baltagi (2008) for a survey on forecasting with panel data, and a more complete list of references.

7 Elliott and Timmermann (2008) discuss and illustrate multi-step forecasting in linear models.

delta method to derive the distribution of multi-period forecast. More recently, Chevillon and Hendry (2005) also apply iteration and the delta method to obtain the asymptotic distribution of iterated multi-step estimators for forecasting.[8] The QR analogous version of this technique could be applied to the $\tau$th conditional quantile estimate based on the $s$-step-ahead iteration of the QR model, and one could use the asymptotic result stated previously along with the delta method to derive the asymptotic distribution of the estimator, $\hat{\delta}(\tau)^s$, and consequently of the conditional quantiles. However, when constructing an $s$-step-ahead forecast for a future response variable using QR models, it is important to consider all the potential trajectories for the corresponding quantiles rather than a fixed $\tau$. Thus, as Koenker and Zhao (1996) suggest, for QR models, a simulation based approach is a suitable tool for $s$-step-ahead forecasting. The approach we propose is based on this method. Let $\hat{Q}_{y_{iT+s}}(\tau|\cdot)$ denote the conditional quantile function of $y_{iT+s}$ given the information up to time $T+s-1$, for a given individual $i$. A draw from the one-step-ahead forecast distribution is given by $\hat{y}_{iT+s} = \hat{Q}_{y_{iT+s}}(U|\cdot)$, where $U$ is a uniformly distributed random variable on $[0, 1]$. Finally, we are able to construct a predictive interval by drawing repetitively from the distribution of the forecasts and taking the selected quantiles of the empirical distribution. A number of other numerical approaches are available for $s$-step-ahead forecasting in the literature. Lin and Granger (1994) describe several numerical techniques for two-step forecasting with nonlinear models.[9] In addition, Peters and Freedman (1985) show that the use of bootstrap to construct multi-period forecasts is more reliable than the asymptotics based on the delta method.

More specifically, quantile regression offers an alternative approach to the construction of $s$-step-ahead prediction intervals. It is important to note that if the parameters of the model were known exactly, the conditional quantile function itself could be used, and the interval

$$[Q_{y_{iT+s}}(\lambda/2), Q_{y_{iT+s}}(1-\lambda/2)],$$

would provide an exact $1-\lambda$ level interval for an $s$-step-ahead forecast. The methods proposed by Koenker and Zhao (1996) and Zhou and Portnoy (1996) suggest the construction of a $1-\lambda$ level interval for an $s$-step-ahead forecast as

$$[\hat{Q}_{y_{iT+s}}(\lambda/2-h_n), \hat{Q}_{y_{iT+s}}(1-\lambda/2+h_n)],$$

where $h_n \to 0$ accounts for parameter uncertainty.[10]

However, an important practical aspect of constructing such forecast interval involves computing $\hat{Q}_{y_{iT+s}}(\cdot)$. This is straightforward in the one-step ahead case, as shown in Eq. (10), but more problematic for $s > 1$. Geweke (1989) discusses a Bayesian approach to construct predictive densities, for the ARCH linear models, by means of Monte Carlo integration with importance sampling. Koenker and Zhao (1996) suggest a related method, based on a simple simulation approach, for construction of out-of-sample prediction for quantile regression ARCH models. A similar approach seems reasonable for the QRPIV problem. Let $Q_{y_{it}}(\tau|y_{it-1}, x_{it})$ denote the conditional quantile function of $y_{it}$, for an individual $i$, given the information up to time $t-1$. A draw from the $s$-step-ahead forecast distribution is given by

$$\begin{aligned}\hat{y}_{iT+s} &= \hat{Q}_{y_{iT+s}}(U|y_{iT+s-1}, x_{iT+s})\\ &= \hat{\eta}_i + \hat{\alpha}(U)\tilde{y}_{iT+s-1} + x'_{iT+s}\hat{\beta}(U),\end{aligned} \tag{11}$$

where $U$ is a uniformly distributed random variable on $[0, 1]$, $x_{iT+s}$ is given, and

$$\tilde{y}_{it} = \begin{cases} y_{it} & \text{if } t \le T,\\ \hat{y}_{it} & \text{if } t > T.\end{cases}$$

Applying (11) recursively we can compute a sample path of forecasts $(\hat{y}_{iT+1}, \hat{y}_{iT+2}, \dots, \hat{y}_{iT+s})'$. Repeatedly applying this procedure, say $R$ times, one can compute the $\lambda/2-h_n$ and $1-\lambda/2+h_n$, quantiles of the empirical distribution of the forecasts and used them to construct the final prediction intervals.[11] It may appear that the use of (11) is computationally prohibitive because it appears to require a quantile regression estimate for each possible realization of $U \in (0, 1)$. However, the entire function $Q_{y_{iT}}(\tau|\cdot)$ is easily computed by standard parametric linear programming techniques, yielding a piecewise constant function on a known grid that is then readily evaluated by the forecasting simulation.

The QRPIV model can also be used to estimate and predict the conditional density function. For instance, let us consider the one-step-ahead forecast. Given the parameter estimates, for an individual $i$, the $\tau$-th conditional quantile function of $y_{iT+1}$, can be estimated by (10). Thus, given a family of estimated conditional quantile functions for an individual $i$, the conditional density of $y_{iT+1}$ at values of the conditioning covariate can be estimated by the difference quotients,

$$\hat{f}_{y_{iT+1}}(\tau|y_{iT}, x_{iT+1}) = \frac{2h_n}{\hat{Q}_{y_{iT+1}}(\tau+h_n|y_{iT}, x_{iT+1}) - \hat{Q}_{y_{iT+1}}(\tau-h_n|y_{iT}, x_{iT+1})}, \tag{12}$$

where $h_n$ is a bandwidth which tends to zero as the sample size tends to infinity. Using the same principle, the $s$-step-ahead conditional density function is immediately available from (12).

Koenker and Bassett (2010) suggest an alternative use of QR methods to construct predictive densities. As described previously, given the estimated quantile regression model (11), it is possible to draw uniform random variables and evaluate the model. Repeated evaluations, say $R$ times, yield a predictive distribution. Thus, density can be estimated using usual kernel methods. This simulation method of producing predictive densities is closely related to the "rearrangement" methods for monotonization of conditional quantile estimates introduced recently by Chernozhukov et al. (2010). Hence, the conditional density forecasts $s$-step-ahead, for the individual $i$, can be constructed based on an ensemble of such forecast paths using the QRPIV model. In practice, the simulation method described previously produces $R$ prediction samples of conditional quantile function of interest for period $T+s$. Thus, one can apply any typical nonparametric density estimator to predict the conditional density function $s$-step-ahead.

A similar method of constructing density forecasts from quantile regressions is proposed by Gaglianone and Lima (2009). They propose a model that generates an optimal individual forecast that is a function of a common factor, consensus forecast, and an idiosyncratic component that depends only on the individual loss function. The conditional density of the $s$-step-ahead forecast can be estimated from the conditional quantile functions, $\hat{f}_{y_{iT+s}}(\tau|\cdot) = (\tau_k - \tau_{k-1})/(\hat{Q}_{y_{iT+s}}(\tau_k|\cdot) - \hat{Q}_{y_{iT+s}}(\tau_{k-1}|\cdot))$, for some appropriately chosen sequence of $\tau$'s. The conditional densities can also be estimated by using the Epanechnikov kernel.

8 When deriving the distribution of the $s$-step-ahead predictor for simple autoregressive and moving average linear regression models, it is customary to impose the assumption of normality of the innovations. In this case, it is relatively straightforward to derive the probability limit of the forecasts at any lead time (see e.g. Box et al. (2008)). For QR we want to avoid to specify the distribution of the innovations.

9 In the QR literature, there is recent work based on numerical methods for predicting with dynamic models. Cai (2010) proposes forecasting methods for obtaining $s$-step-ahead predictive quantiles based on numerical and Monte Carlo methods; and Lee and Yang (2008) suggest a bootstrap aggregating for multi-step forecast horizons for nonlinear models.

10 See Zhou and Portnoy (1996) Theorem 2.1 for a Bahadur representation, and Corollary 2.1 for asymptotic normality and confidence interval.

11 We discuss the choice of $h_n$ in the next section.

**Table 1**
Location-shift model: bias and RMSE of estimators for Normal ($G$) and $t_3$ distributions ($T = 10$ and $N = 50$).

| | | | WG | OLS-IV | QRP | | | QRPIV | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\tau = 0.25$ | $\tau = 0.5$ | $\tau = 0.75$ | $\tau = 0.25$ | $\tau = 0.5$ | $\tau = 0.75$ |
| $G$ | $\alpha = 0.5$ | Bias | −0.0982 | −0.0070 | −0.1044 | −0.0976 | −0.0953 | −0.0190 | −0.0013 | 0.0097 |
| | | RMSE | 0.103 | 0.082 | 0.113 | 0.109 | 0.107 | 0.093 | 0.089 | 0.088 |
| | $\beta = 0.7$ | Bias | 0.0349 | −0.0006 | −0.0019 | 0.0325 | 0.0683 | −0.0158 | −0.0015 | 0.0178 |
| | | RMSE | 0.055 | 0.065 | 0.063 | 0.057 | 0.091 | 0.075 | 0.069 | 0.069 |
| $t_3$ | $\alpha = 0.5$ | Bias | −0.1432 | 0.0107 | −0.0969 | −0.1018 | −0.0923 | −0.0155 | −0.0055 | 0.0056 |
| | | RMSE | 0.153 | 0.151 | 0.109 | 0.114 | 0.106 | 0.098 | 0.095 | 0.094 |
| | $\beta = 0.7$ | Bias | 0.0635 | 0.0115 | −0.0134 | 0.0364 | 0.0779 | −0.0144 | −0.0018 | 0.0166 |
| | | RMSE | 0.106 | 0.117 | 0.074 | 0.065 | 0.111 | 0.083 | 0.073 | 0.092 |

## 4. Monte Carlo simulation

We use simulation experiments to assess the finite sample performance of the quantile regression estimator discussed in the previous sections. Two simple versions of the basic model in Eq. (1) are considered in the simulation experiments. In the first, the exogenous covariate, $x_{it}$, exerts a pure location shift effect. In the second, $x_{it}$ exerts both location and scale effects. In the former case the response $y_{it}$ is generated by the model,

$$y_{it} = \eta_i + \alpha y_{it-1} + \beta x_{it} + u_{it}$$

while in the latter case,

$$y_{it} = \eta_i + \alpha y_{it-1} + \beta x_{it} + (\gamma x_{it})u_{it}.$$

We employ two different schemes to generate the disturbances $u_{it}$. Under Scheme 1, we generate $u_{it}$ as a $N(0, \sigma_u^2)$, and under Scheme 2 we generate $u_{it}$ as a $t$-distribution with 3 degrees of freedom. The regressor $x_{it}$ is generated according to $x_{it} = \mu_i + \zeta_{it}$, where $\zeta_{it}$ follows an ARMA(1, 1) process,

$$(1 - \phi L)\zeta_{it} = \epsilon_{it} + \theta\epsilon_{it-1},$$

and $\epsilon_{it}$ follows the same distribution as $u_{it}$, that is, normal distribution and $t_3$ for Schemes 1 and 2, respectively. In all cases we set $\zeta_{i,-50} = 0$ and generate $\zeta_{it}$ for $t = -49, -48, \ldots, T$. This ensures that the results are not unduly influenced by the initial values of the $x_{it}$ process. In generating $y_{it}$ we also set $y_{i,-50} = 0$ and discard the first 50 observations, using the observations $t = 0$ through $T$ for estimation. The fixed effects, $\mu_i$ and $\eta_i$, are generated as

$$\mu_i = e_{1i} + T^{-1}\sum_{t=1}^{T}\epsilon_{it}, \quad e_{1i} \sim N(0, \sigma_{e_1}^2);$$

$$\eta_i = e_{2i} + T^{-1}\sum_{t=1}^{T}x_{it}, \quad e_{2i} \sim N(0, \sigma_{e_2}^2).$$

In the simulations, we consider $T = \{10, 20, 50\}$ and $N = \{50, 100\}$. We set the number of replications to 2000, and consider the following values for the remaining parameters: $(\alpha, \beta) = (0.5, 0.7)$, $\phi = 0.7$, $\theta = 0.2$, $\gamma = 0.5$, $\sigma_u^2 = \sigma_{e_1}^2 = \sigma_{e_2}^2 = 1$.

### 4.1. Bias and RMSE

In this section, the bias and root mean squared error (RMSE) of different estimators are presented. We evaluate four different estimators, the within group (WG), the OLS instrumental variables (OLS-IV), the fixed effects panel quantile regression (QRP) proposed by Koenker (2004), and the fixed effects quantile regression dynamic panel instrumental variables (QRPIV) proposed in this paper. The quantile regression based estimators are analyzed for quantiles $\tau = (0.25, 0.5, 0.75)$. For the OLS-IV and QRPIV we considered two different instruments, $y_{it-2}$ and $x_{it-1}$; the results are similar in both cases, and we simply present results for the $x_{it-1}$ case. We also consider different sample sizes in the experiments. However, due to space limitations we only report results for $T = 10$, $N = 50$. The results for the other sample sizes are similar. Tables 1 and 2 present bias and RMSE results for estimates of the autoregression coefficient, $\alpha$, and the exogenous variable coefficient, $\beta$, for the location-shift and the location-scale-shift models, respectively.

Table 1 collects results for the location-shift model and shows that, in the Gaussian case ($G$) the autoregressive coefficient is biased downward for the WG case, but the OLS-IV is approximately unbiased. Likewise, in the presence of lagged variables, $\hat{\alpha}$ is biased downward for QRP, and the QRPIV estimator is able to largely eliminate the bias. Table 1 also reveals that $\hat{\beta}$ is slightly biased in the WG and QRP cases, and unbiased for both OLS-IV and QRPIV. The results regarding the $t_3$-distribution show that the autoregressive estimates of WG and QRP are biased downward, and the QRPIV and OLS-IV are approximately unbiased estimators for both coefficients. Regarding the RMSE, in the Gaussian case, the OLS-based estimators perform better than the respective quantile regression estimators. In contrast, for the heavier-tailed $t_3$ distribution of innovations, the RMSE results from the quantile estimators are smaller than their respective OLS based estimators.

The results for the location-scale-shift model are presented in Table 2. They are qualitatively similar to those in Table 1, where QRPIV is approximately unbiased, with improved precision relative to OLS-IV, in the $t_3$ case. Table 2 shows that in both distributional cases, the WG and QRP estimators are biased downward and the OLS-IV and QRPIV are approximately unbiased. The RMSE's present the same features as in the previous case as well, where under non-Gaussian heavier-tailed conditions, $t_3$, the quantile regression estimators perform better than the least squares based estimators.

Finally, we examine the bias of the autoregressive estimator when $\alpha$ varies. We estimate the location model for the Normal and $t_3$ distribution cases varying $\alpha \in (0, 1)$. The results are essentially the same and presented in Fig. 1. They show that for values of $\alpha$ very close to zero, there is a very small positive bias, which disappears for most values of $\alpha$. However, as the autoregression coefficient increases toward the unit, the bias might be large. This result is in line with the literature on dynamic panel data for conditional mean, where for $\alpha \approx 1$ the instruments from lagged variables tend to be weak.[12]

### 4.2. Prediction

First we evaluate the performance of the prediction interval in terms of empirical coverage probabilities and empirical lengths. We set the nominal level equal to 0.90 (i.e. $\lambda = 0.9$), $N = 50$, and vary the time series dimension over $T = \{10, 20, 50\}$. We follow the approach proposed by Zhou and Portnoy (1996) and set the constant $h_n = z_\alpha\sqrt{\tilde{x}'D^{-1}\tilde{x}\tau(1-\tau)}/\sqrt{NT}$, for fixed $\tilde{x}$, with $D = (NT)^{-1}\sum_{t=1}^{T}\sum_{i=1}^{N}\tilde{x}_{it}\tilde{x}_{it}'$, and $\tau = 0.5$. We compute predictive intervals for $(y_{iT}, y_{iT+1}, y_{iT+2}, y_{iT+3})$. Eq. (11) is computed recursively from the location and location-scale models. Moreover, the number of draws from the uniform distribution is $R = 500$

12 See, e.g., Arellano and Bover (1995) and Ahn and Schmidt (1995) for more details.

**Table 2**
Location-scale-shift model: bias and RMSE of estimators for Normal ($G$) and $t_3$ distributions ($T = 10$ and $N = 50$).

| | | | WG | OLS-IV | QRP | | | QRPIV | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\tau = 0.25$ | $\tau = 0.5$ | $\tau = 0.75$ | $\tau = 0.25$ | $\tau = 0.5$ | $\tau = 0.75$ |
| $G$ | $\alpha = 0.5$ | Bias | −0.0967 | 0.0019 | −0.0939 | −0.0959 | −0.0987 | −0.0073 | −0.0003 | 0.0069 |
| | | RMSE | 0.099 | 0.078 | 0.104 | 0.106 | 0.109 | 0.083 | 0.083 | 0.086 |
| | $\beta = 0.7$ | Bias | 0.0454 | 0.0003 | −0.0337 | 0.0180 | 0.0699 | −0.0053 | −0.0018 | 0.0172 |
| | | RMSE | 0.077 | 0.080 | 0.081 | 0.085 | 0.091 | 0.082 | 0.084 | 0.086 |
| $t_3$ | $\alpha = 0.5$ | Bias | −0.1244 | 0.0173 | −0.0881 | −0.0894 | −0.0905 | −0.0152 | −0.0044 | 0.0042 |
| | | RMSE | 0.139 | 0.132 | 0.103 | 0.107 | 0.109 | 0.079 | 0.085 | 0.086 |
| | $\beta = 0.7$ | Bias | 0.0727 | 0.0104 | −0.0415 | 0.0238 | 0.0905 | −0.0186 | 0.0011 | 0.0125 |
| | | RMSE | 0.132 | 0.153 | 0.081 | 0.063 | 0.119 | 0.087 | 0.088 | 0.096 |

**Table 3**
Location-shift model: empirical coverage probabilities and length for predictive interval with nominal level 0.90.

| | | Normal | | | | $t_3$ | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | $y_T$ | $y_{T+1}$ | $y_{T+2}$ | $y_{T+3}$ | $y_T$ | $y_{T+1}$ | $y_{T+2}$ | $y_{T+3}$ |
| CP | $T = 10$ | 0.929 | 0.863 | 0.832 | 0.828 | 0.907 | 0.852 | 0.844 | 0.829 |
| | $T = 20$ | 0.903 | 0.875 | 0.861 | 0.841 | 0.879 | 0.861 | 0.857 | 0.837 |
| | $T = 50$ | 0.896 | 0.888 | 0.858 | 0.849 | 0.887 | 0.879 | 0.858 | 0.851 |
| Length | $T = 10$ | 3.076 | 3.074 | 4.608 | 5.395 | 4.877 | 4.870 | 7.353 | 8.698 |
| | $T = 20$ | 2.920 | 3.657 | 4.374 | 5.106 | 4.192 | 4.188 | 6.312 | 7.422 |
| | $T = 50$ | 2.927 | 2.927 | 4.093 | 4.830 | 4.058 | 4.062 | 6.127 | 7.088 |

**Table 4**
Location-scale-shift model: empirical coverage probabilities and length for predictive interval with nominal level 0.90.

| | | Normal | | | | $t_3$ | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | $y_T$ | $y_{T+1}$ | $y_{T+2}$ | $y_{T+3}$ | $y_T$ | $y_{T+1}$ | $y_{T+2}$ | $y_{T+3}$ |
| CP | $T = 10$ | 0.915 | 0.854 | 0.849 | 0.832 | 0.922 | 0.825 | 0.848 | 0.848 |
| | $T = 20$ | 0.866 | 0.858 | 0.843 | 0.834 | 0.929 | 0.865 | 0.879 | 0.853 |
| | $T = 50$ | 0.892 | 0.864 | 0.856 | 0.850 | 0.918 | 0.873 | 0.874 | 0.860 |
| Length | $T = 10$ | 5.963 | 5.967 | 8.972 | 10.603 | 10.620 | 10.611 | 15.872 | 17.801 |
| | $T = 20$ | 5.239 | 5.237 | 7.872 | 9.268 | 10.939 | 10.799 | 13.331 | 16.627 |
| | $T = 50$ | 5.140 | 5.138 | 7.683 | 8.969 | 8.471 | 9.671 | 13.761 | 14.579 |

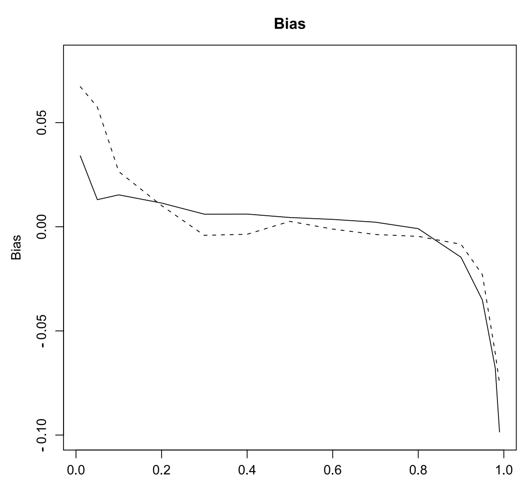


**Fig. 1.** Bias autoregressive coefficient. The solid line shows the bias for the Normal distribution case, the dashed line indicates the bias for the $t_3$ distribution case.

and $x_{iT+1}$ is the last observation in the sample. Since there are $N$ intervals, we present results of the average over the individuals, but the results do not change for a fixed individual $i$. The results for both distributions are summarized in Tables 3 and 4 for location and location-scale cases respectively.

Table 3 presents results for empirical coverage probabilities and length for prediction interval with nominal level 0.90 for the location case. The simulations results show that, for small time series and in-sample prediction, the interval is slightly conservative on coverage probabilities. However, the conservativeness disappears when sample size grows. In addition, results present evidence that, for a given sample size, the proposed method to compute predictive interval tends to underestimate the coverage levels for longer forecasting horizons. However, this phenomenon disappears for larger sample sizes. As the time series dimension increases, the coverages probability improve with respect to the nominal level 0.9. The results regarding the empirical length of the prediction interval show that, holding a time series sample size constant, the interval is wider for longer horizons of prediction, but the length decreases when we fix the forecast horizon and increase the sample size.

In Table 4 we present results for the location-scale model. The results are qualitatively similar to those in Table 3. The empirical coverage probability for both Normal and $t_3$ cases are close to the nominal 90%. For all out-of-sample cases, the coverage increases with the sample size. Regarding the length of the interval, as in Table 3, it decreases as sample size increases, and increases as prediction horizon increases.

Second, we study the finite sample performance of the predictive conditional density method. We estimate the one-step-ahead density and compare it with the true underline density function from the location model. In this simulation we use a simple Gaussian kernel smooth technique with default bandwidth in *R*. Fig. 2 presents results for four random subjects in one run of simulation. The solid (black) line is the true conditional density, the dashed (red) line indicates the predictive one-step-ahead conditional density estimates by the kernel method. Fig. 2 shows that the conditional density estimator closely follows the true density function and thus presents a good performance in terms of predictive conditional density function. We also estimate the models using the Epanechnikov kernel, as suggested by Gaglianone and Lima (2009), the results remain essentially the same.

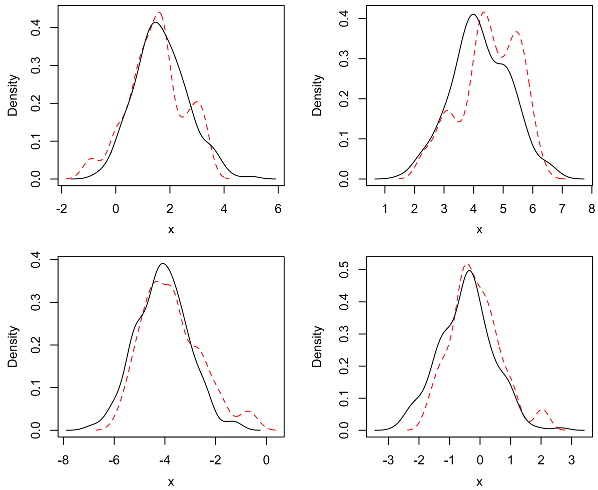

**Fig. 2.** Predictive density function. The solid (black) line shows the true conditional density, the dashed (red) line indicates the one-step-ahead predictive conditional density estimates using the Gaussian kernel method.

## 5. Application

Aggregate output is probably the most popular macroeconomic variable when it comes to forecasting. We illustrate the proposed approach with an application to forecasting gross domestic product (GDP) growth rates for 18 OECD (Organization for Economic Cooperation and Development) countries. Panel data techniques have been frequently applied to forecast multi-country output growth rates, and the use of such models to generate a forecast for GDP growth rates is important because it is crucial to take into account time invariant unobserved heterogeneity across countries (fixed effects). Pure cross-section studies cannot control for the unobserved country effects, whereas pure time-series studies cannot control for unobserved changes occurring along the time. In addition, the use of QR methods is important to capture heterogeneity in the model, to give more flexibility with no distributional assumptions on the innovations, and to produce robust estimates.

There are several studies on forecasting growth rates for OECD countries using dynamic panel data. In a study of real gross national product growth rates of nine OECD countries, Garcia-Ferrer et al. (1987) showed that pooled estimates of an dynamic model with leading indicator variables provided superior forecasting results. Hoogstrate et al. (2000) investigate the improvement of forecasting performance using pooling techniques instead of single country forecasts for $N$ fixed and $T$ large. They use a set of dynamic regression equations and GDP growth rates of 18 OECD countries. Gavin and Theodorou (2005) use forecasting criteria to examine the macroeconomic behavior of 15 OECD countries. They find that the small set of variables and a simple VAR common model strongly support the hypothesis that many industrialized countries have similar macroeconomic dynamics.[13] There is also a literature on analyzing economic growth models using QR techniques (see e.g. Crespo-Cuaresma et al. (2009), Canarella and Pollard (2004), and Mello and Novo (2002)). Moreover, there is recent work on forecasting economic growth rates with time-series QR models (see e.g. Cai (2007, 2010)). However, to the best of our knowledge, there is a lack of literature on forecasting with panel data QR models.

The data used in this analysis come from the International Monetary Funds International Financial Statistics Data Base and contain the following variables: (i) GDP; (ii) real stock prices; (iii) a GDP deflator. We use data on the following 18 countries: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, The Netherlands, Norway, Spain, Sweden, Switzerland, United Kingdom, and United States. The longest series contain annual data from 1948 to 2008, but data points are missing for some countries, making the panel unbalanced.

We construct one-step-ahead predictions using a simple autoregressive dynamic model, and an alternative model including leading indicator (LI) variables. The following dynamic panel quantile regression model was employed to generate the output growth rates forecasts:

$$Q_{y_{it}}(\tau|\mathcal{F}_{it-1}) = \eta_i + \alpha(\tau)y_{it-1} + \beta_1(\tau)SR_{it-1} + \beta_2(\tau)SR_{it-2}, \tag{13}$$

where $y_{it}$ denotes the first difference of the logarithm of the real output, and $SR_{it}$ denotes the first difference of the log of a stock price index. We also estimate the least squares version of (13) for comparison reasons. We use $y_{t-2}$ as instruments for both models.

First, we estimate the dynamic autoregressive model without LI variables ($\beta_1 = \beta_2 = 0$). We compare the estimates for the autoregressive coefficient, $\alpha$, for both QRPIV, with $\tau = 0.5$, and OLS-IV. The results are 0.403 (0.15) for QRPIV, and 0.386 (0.07) for OLS-IV, with standard errors inside parentheses. These results show similar point estimates for both conditional mean and median models. Moreover, we construct prediction intervals for $y_{iT+1}$ (2009) with nominal level 0.90. For QRPIV, we use the simulation technique described in Section 3, and for OLS-IV we assume Gaussian innovations, for simplicity and such that we can compare later the fit of the predictive densities. The results for lower bound (LB), upper bound (UB), and length (LG) of the intervals are presented in Table 5 (Model 1). When comparing the length, we observe a larger variation in QRPIV than OLS-IV, with Finland and Ireland presenting the largest lengths. For these two countries the growth rate of the real GDP ranges from negative values to strong positive growth rates. A preliminary analysis of the QRPIV intervals shows that the countries with better prognosis in terms of economic growth are France, Norway, and Spain. For all the other countries, the one-step-ahead intervals for output growth rate present negative LB, for both QRPIV and OLS-IV cases. In addition, note that the OLS-IV delivers negative estimates for all LB, and they are, in general, more negative than QRPIV.

Using these estimators we also construct densities for the one-step-ahead forecast. For QRPIV we estimate the conditional density functions using the simulation method described in Section 3, and for OLS-IV we make use of the normality assumption.[14] The results are presented in Fig. 3, and show evidence of similar estimates of the peak of the distributions for almost all countries, with exception of Belgium, Ireland, Italy, Norway, Spain, and Sweden. Although, one important distinction is that the QRPIV model has often a higher peak at center of the distribution than OLS-IV. This is due to the normality assumption for the latter. Another important difference between the two methods is that for several countries the QRPIV densities are asymmetric, while this feature cannot be captured with OLS-IV. Moreover, from the QRPIV results it can be seen that Finland and Japan have considerable amount of probability mass on their left tails in negative regions. Thus, according to these estimates, these countries were likely to face a recession.

13 In related work, Fok et al. (2005) use a panel of 48 US states to show that forecasts of aggregates, as total output, can be improved by considering panel models of disaggregated series.

14 There are several methods to compute density forecasting available in the literature, see for instance Tay and Wallis (2000). We impose normality for simplicity and comparison reasons. We compute standard errors for the individual specific effects as in Arellano (2003).

**Table 5**
One-step-ahead predictive intervals for OECD growth rate of the real GDP.

| | Model 1 | | | | | | Model 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | QRPIV | | | OLS-IV | | | QRPIV | | | OLS-IV | | |
| | LB | UB | LG | LB | UB | LG | LB | UB | LG | LB | UB | LG |
| Australia | −0.001 | 0.075 | 0.076 | −0.006 | 0.069 | 0.075 | −0.022 | 0.052 | 0.074 | −0.003 | 0.060 | 0.063 |
| Austria | −0.005 | 0.052 | 0.058 | −0.013 | 0.060 | 0.074 | −0.036 | 0.065 | 0.100 | −0.012 | 0.053 | 0.065 |
| Belgium | −0.004 | 0.055 | 0.060 | −0.014 | 0.059 | 0.073 | −0.032 | 0.043 | 0.075 | −0.014 | 0.050 | 0.064 |
| Canada | −0.014 | 0.063 | 0.077 | −0.015 | 0.060 | 0.075 | −0.019 | 0.027 | 0.046 | −0.005 | 0.055 | 0.060 |
| Denmark | −0.016 | 0.055 | 0.071 | −0.032 | 0.040 | 0.073 | −0.048 | 0.004 | 0.053 | −0.028 | 0.035 | 0.063 |
| Finland | −0.026 | 0.087 | 0.113 | −0.015 | 0.062 | 0.077 | −0.031 | 0.049 | 0.080 | −0.011 | 0.052 | 0.064 |
| France | 0.011 | 0.053 | 0.042 | −0.014 | 0.060 | 0.074 | −0.023 | 0.039 | 0.063 | −0.010 | 0.053 | 0.063 |
| Germany | −0.008 | 0.074 | 0.082 | −0.020 | 0.055 | 0.075 | −0.029 | 0.056 | 0.085 | −0.018 | 0.046 | 0.064 |
| Ireland | −0.007 | 0.102 | 0.110 | −0.023 | 0.052 | 0.076 | −0.062 | 0.027 | 0.089 | −0.014 | 0.049 | 0.063 |
| Italy | −0.011 | 0.039 | 0.050 | −0.030 | 0.045 | 0.075 | −0.048 | 0.032 | 0.080 | −0.024 | 0.039 | 0.063 |
| Japan | −0.019 | 0.068 | 0.086 | −0.014 | 0.061 | 0.075 | −0.055 | 0.039 | 0.094 | −0.004 | 0.061 | 0.065 |
| Netherlands | 0.000 | 0.067 | 0.068 | −0.007 | 0.065 | 0.073 | −0.026 | 0.049 | 0.075 | −0.009 | 0.056 | 0.065 |
| Norway | 0.003 | 0.054 | 0.051 | −0.009 | 0.065 | 0.074 | −0.027 | 0.042 | 0.069 | −0.007 | 0.057 | 0.064 |
| Spain | 0.003 | 0.071 | 0.068 | −0.008 | 0.065 | 0.074 | −0.021 | 0.071 | 0.091 | −0.005 | 0.059 | 0.064 |
| Sweden | −0.008 | 0.051 | 0.058 | −0.021 | 0.053 | 0.074 | −0.042 | 0.021 | 0.063 | −0.017 | 0.046 | 0.063 |
| Switzerland | −0.011 | 0.057 | 0.068 | −0.020 | 0.054 | 0.075 | −0.020 | 0.043 | 0.063 | −0.021 | 0.042 | 0.063 |
| United Kingdom | −0.011 | 0.058 | 0.069 | −0.018 | 0.055 | 0.073 | −0.002 | 0.063 | 0.065 | −0.004 | 0.059 | 0.063 |
| United States | −0.009 | 0.069 | 0.079 | −0.015 | 0.058 | 0.074 | −0.029 | 0.055 | 0.083 | −0.011 | 0.055 | 0.066 |

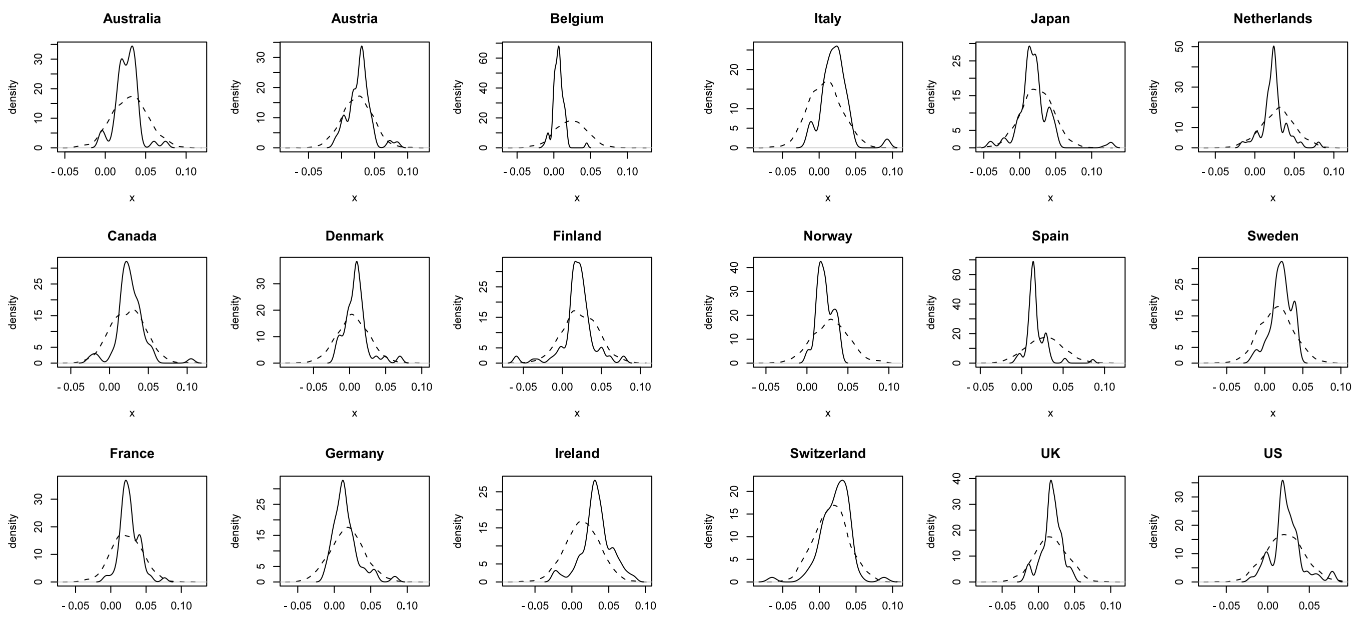


**Fig. 3.** One-step-ahead predictive density functions for model 1. The solid line is the one-step-ahead predictive density based on the quantile regression, the dashed line indicates the one-step-ahead predictive density estimate using the mean regression method.

In a second round of estimations, we include $SR_{it}$ as a LI variable, and estimate Eq. (13). First, we compare the autoregressive coefficient. For median QRPIV the coefficient is 0.526 (0.16) and for OLS-IV we have 0.185 (0.07). Thus, the results show a large difference between these coefficients, with QRPIV increasing and OLS-IV decreasing. Again, we focus on one-step-ahead forecast and construct prediction intervals and density functions for each country in the sample. The data for stock price index has a considerable number of missing values. Denmark has data only from 1996 to 2008, and Switzerland has data from 1989 to 2008. In addition, for UK we only have data from 1958 to 1998, such that forecasting one-step-ahead has a different interpretation than for the other countries.[15] Table 5 (Model 2) presents the one-step-ahead prediction interval estimates for LB, UB, and LG.

Relative to the previous model, without controlling for stock price index variable, on one hand, the forecasts for the QRIV are more pessimistic with all the countries presenting a negative lower bound and zero into the interval. On the other hand, in general, the intervals for OLS-IV are more optimistic with less negative LB. For the QRPIV, the lengths are similar, with exception of Austria, Japan, and Spain, which present a considerably wider length. Regarding the OLS-IV, the lengths are narrower, and once again, present less variation than QRPIV. Another important feature one can observe is that, in general, the intervals have smaller UB GDP growth rate for both cases.

We also construct predictive density functions for this case. The results are presented in Fig. 4, and show that the models produce similar densities for Australia, Austria, Netherlands, Norway, Switzerland, and UK. However, for the remaining countries, the results are quite different, and the most prominent features are the higher peaks for QRPIV, and that densities from QRPIV model are asymmetric with considerable more mass on the left tail, thus,

15 Due to the missing data problem, we do not interpret results for the United Kingdom.

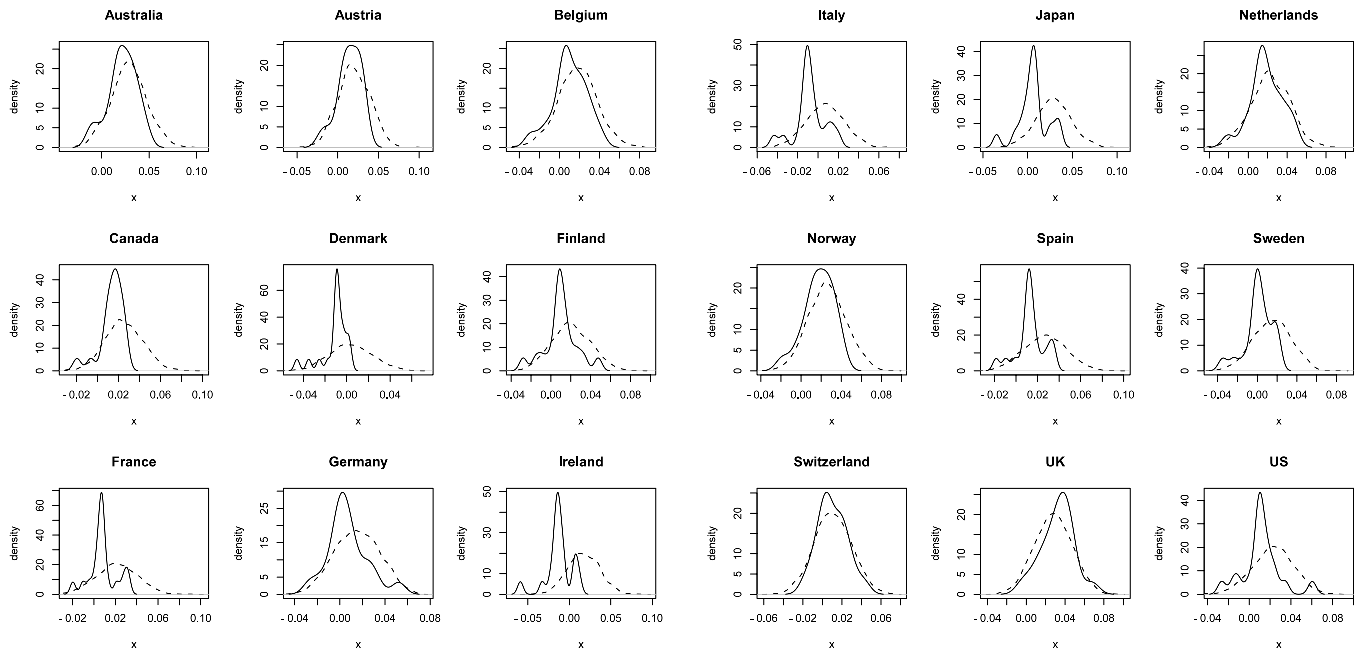


**Fig. 4.** One-step-ahead predictive density functions for model 2. The solid line is the one-step-ahead predictive density based on the quantile regression, the dashed line indicates the one-step-ahead predictive density estimate using the mean regression method.

forecasting smaller growth rates than OLS-IV models. It is possible to observe that for QRPIV, relative to the previous model, the left tail presents much more mass for most of the densities, and the mean is shifted to the left. Thus, most predictive densities indicate a recession with more probability relative to the previous model. An interesting phenomenon happened with the QRPIV densities of Belgium, Canada, Denmark, and Norway, which became negatively skewed in the second round of estimation.

This exercise illustrates forecasting with QRPIV, and compares the results with a simple OLS-IV. The results suggest evidence of important differences. In particular the density estimates for the model with LI variables differ considerably. In addition, the results show that presenting density forecasts is an important complement to interval forecasting, since the former conveys additional important information about the distribution of probability mass. Furthermore, using QR models for this forecasting requires no explicit distributional assumptions on the densities of the innovations.

## 6. Conclusion

This paper studies estimation, inference, and prediction in a quantile regression dynamic panel model with fixed effects. Standard approaches to estimate dynamic panel models with FE are typically biased in the presence of lagged dependent variables as regressors. To reduce the dynamic bias in the quantile regression FE estimator we suggest the use of the IV quantile regression method of Chernozhukov and Hansen (2006) along with lagged regressors as instruments. In addition, we describe how to construct prediction intervals and conditional density functions using the proposed model. Monte Carlo simulations show evidence that the quantile regression IV approach for dynamic panel data sharply reduces the dynamic bias and turns out to be especially advantageous when innovations are heavy-tailed. Moreover, the empirical levels for predictive intervals approximate fairly well the nominal or theoretical levels. Finally, we illustrate the methods with an application to forecasting output growth rates for 18 OECD countries. We compute one-step-ahead prediction intervals and density functions.

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## Appendix

The next two lemmas help in the derivation of the results.

**Lemma 1.** *Given assumptions* A1–A6, $(\eta, \beta(\tau), \alpha(\tau))$ *uniquely solves the equations* $E[\upsilon\psi(y - Z\eta - y_{-1}\alpha - X\beta)\check{X}(\tau)] = 0$ *over* $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$.

**Proof.** This is a simple application of Theorem 2 of Chernozhukov and Hansen (2006), which shows that $(\eta, \alpha(\tau), \beta(\tau))$ uniquely solves the limit problem for each $\tau$, that is, $\eta(\alpha^*(\tau)) = \eta, \alpha^*(\tau) = \alpha(\tau)$, and $\beta(\alpha^*(\tau), \tau) = \beta(\tau)$. Let $\psi(u) := (\tau - I(u < 0))$, and

$$H(\eta, \alpha, \beta, \tau) := \frac{\partial}{\partial(\eta, \alpha, \beta)} E[\upsilon\psi(y - Z\eta - y_{-1}\alpha(\tau) - X\beta(\tau))\check{X}(\tau)].$$

By assumption A2, $H(\eta, \alpha, \beta, \tau)$ is continuous in $(\eta, \alpha, \beta)$ and full rank, uniformly over $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$. Moreover, by A2, the image of the set $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$ under the mapping $(\eta, \alpha, \beta) \to \Pi(\eta, \alpha, \beta, \tau)$ is simply connected. As in Chernozhukov and Hansen (2005), the application of Hadamard's global univalence theorem for general metric spaces yields that the mapping $\Pi(\cdot, \cdot, \cdot, \tau)$ is a homeomorphism (one-to-one) between $(\mathscr{E} \times \mathscr{A} \times \mathscr{B})$ and $\Pi(\mathscr{E}, \mathscr{A}, \mathscr{B}, \tau)$, the image of $\mathscr{E} \times \mathscr{A} \times \mathscr{B}$ under $\Pi(\cdot, \cdot, \cdot, \tau)$. Since $(\eta, \alpha, \beta) = (\eta, \alpha(\tau), \beta(\tau))$ solves the equation

$\Pi(\eta, \alpha, \beta, \tau) = 0$; and thus it is the only solution in $(\mathscr{E} \times \mathscr{A} \times \mathscr{B})$. This argument is valid for $\tau \in \mathscr{T}$. The remaining of the argument is the same as in Chernozhukov and Hansen (2006), and $(\eta(\alpha(\tau)) = \eta, \alpha(\tau), \beta(\alpha(\tau), \tau) = \beta(\tau))$ it is the only such solution for the problem. $\square$

**Lemma 2.** *Let $\xi_{it}(\tau_k) = \eta z_{it} + \alpha(\tau_k) y_{it-1} + \beta(\tau_k) x_{it} + \gamma(\tau_k) w_{it}$, and $u_{it}(\tau_k) = y_{it} - \xi_{it}(\tau_k)$. Let $\vartheta := (\eta, \alpha, \beta, \gamma)$ be a parameter vector in $\mathscr{V} := \mathscr{E} \times \mathscr{A} \times \mathscr{B} \times \mathscr{G}$. Let*

$$\delta = \begin{pmatrix} \delta_\eta \\ \delta_\alpha \\ \delta_\beta \\ \delta_\gamma \end{pmatrix} = \begin{pmatrix} \sqrt{T}(\hat{\eta} - \eta) \\ \sqrt{NT}(\hat{\alpha}(\tau_k) - \alpha(\tau_k)) \\ \sqrt{NT}(\hat{\beta}(\tau_k) - \beta(\tau_k)) \\ \sqrt{NT}(\hat{\gamma}(\tau_k) - \gamma(\tau_k)), \end{pmatrix}.$$

*Under conditions* A1–A6,

$$\sup_{\vartheta \in \mathscr{V}} (NT)^{-1} \left| \sum_k \sum_i \sum_t \left[ \rho_\tau \left( u_{it}(\tau_k) - \frac{z_{it}\delta_\eta}{\sqrt{T}} - \frac{y_{it-1}\delta_\alpha}{\sqrt{NT}} - \frac{x_{it}\delta_\beta}{\sqrt{NT}} - \frac{w_{it}\delta_\gamma}{\sqrt{NT}} \right) - \rho_\tau(u_{it}(\tau_k)) - E\left[ \rho_\tau \left( u_{it}(\tau_k) - \frac{z_{it}\delta_\eta}{\sqrt{T}} - \frac{y_{it-1}\delta_\alpha}{\sqrt{NT}} - \frac{x_{it}\delta_\beta}{\sqrt{NT}} - \frac{w_{it}\delta_\gamma}{\sqrt{NT}} \right) - \rho_\tau(u_{it}(\tau_k)) \right] \right] \right| = o_p(1). \quad (14)$$

**Proof.** With some abuse of notation let $\tilde{x}_{it} = (y_{it-1}, x_{it}, w_{it})$, $\tilde{\beta} = (\alpha, \beta, \gamma)$, and $\vartheta = (\tilde{\beta}, \eta)$. Let $\|\cdot\|$ denote the Euclidean norm. It is sufficient to show that for any $\epsilon > 0$

$$P\left( \sup_{\vartheta \in \mathscr{V}} (NT)^{-1} \left| \sum_k \sum_i \sum_t \left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) - E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) \right] \right] \right| > \epsilon \right) \to 0. \quad (15)$$

For each $k$, consider a partition the parameter space $\Gamma = \mathscr{V}$ into $K_N$ disjoint parts $\Gamma_1, \Gamma_2, \ldots, \Gamma_{K_N}$, such that the diameter of each part is less than $q_{NT} = \frac{\epsilon N^a}{12KC_1T}$. Let $p_N$ be the dimension of $\vartheta$, $p_N = O(N)$, and then $K_N \leq (2\sqrt{p_N}/q_{NT} + 1)^{p_N}$ (c.f. Wei and He (2006)). Let $\zeta_i \in \Gamma_i, i = 1, \ldots, K_N$ be fixed points. Then the left-hand side of (15) can be bounded by $P_1 + P_2$, where

$$P_1 = P\left( \max_{1 \leq q \leq K_N} \sup_{\vartheta \in \Gamma_q} (NT)^{-1} \times \left| \sum_k \sum_i \sum_t \left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) - \rho\left( u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}} \right) + \rho(u_{it}) - E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) \right] + E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}} \right) - \rho(u_{it}) \right] \right] \right| > \epsilon/2 \right)$$

and

$$P_2 = P\left( \max_{1 \leq q \leq K_N} (NT)^{-1} \left| \sum_k \sum_i \sum_t \left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\zeta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_\eta}{\sqrt{T}} \right) + \rho(u_{it}) - E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}} \right) - \rho(u_{it}) \right] \right] \right| > \epsilon/2 \right).$$

Therefore, it suffices to show that both $P_1$ and $P_2$ are $o_p(1)$. Noting that $|\rho(x+y) - \rho(x)| \leq 2|y|$, we have

$$\begin{aligned}
&\max_{1 \leq q \leq K_N} \sup_{\vartheta \in \Gamma_q} (NT)^{-1} \left| \sum_k \sum_i \sum_t \left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) - \rho\left( u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}} \right) + \rho(u_{it}) \right.\right. \\
&\quad \left.\left. - E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\delta_{\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\delta_\eta}{\sqrt{T}} \right) - \rho(u_{it}) \right] + E\left[ \rho \left( u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}} \right) - \rho(u_{it}) \right] \right] \right| \\
&\leq \max_{1 \leq q \leq K_N} \sup_{\vartheta \in \Gamma_q} (NT)^{-1} \times \left[ 4 \left| \sum_k \sum_i \sum_t \left[ \frac{\tilde{x}_{it}}{\sqrt{NT}} (\delta_{\tilde{\beta}} - \zeta_{q\tilde{\beta}}) - \frac{z_{it}}{\sqrt{T}} (\delta_\eta - \zeta_{q\eta}) \right] \right| + 2 \left| \sum_k \sum_i \sum_t E\left[ \frac{\tilde{x}_{it}}{\sqrt{NT}} (\delta_{\tilde{\beta}} - \zeta_{q\tilde{\beta}}) - \frac{z_{it}}{\sqrt{T}} (\delta_\eta - \zeta_{q\eta}) \right] \right| \right] \\
&\leq \max_{1 \leq q \leq K_N} \sup_{\vartheta \in \Gamma_q} (NT)^{-1} 4 \sum_k \sum_i \sum_t \left[ \frac{1}{\sqrt{NT}} \|\tilde{x}_{it}\| \, \|(\delta_{\tilde{\beta}} - \zeta_{q\tilde{\beta}})\| - \frac{1}{\sqrt{T}} \|z_{it}\| \, \|(\delta_\eta - \zeta_{q\eta})\| \right] \\
&\quad + 2 \sum_k \sum_i \sum_t E\left[ \frac{1}{\sqrt{NT}} \|\tilde{x}_{it}\| \, \|(\delta_{\tilde{\beta}} - \zeta_{q\tilde{\beta}})\| - \frac{1}{\sqrt{T}} \|z_{it}\| \, \|(\delta_\eta - \zeta_{q\eta})\| \right] \\
&\leq [4q_{NT} + 2E[q_{NT}]] \leq \frac{\epsilon}{2},
\end{aligned}$$

where the last equality follows from assumptions A5–A6, and implies $P_1 = o_p(1)$. It remains to show that $P_2 = o_p(1)$. If we write $m_{it} = \sum_k [\rho(u_{it} - \tilde{x}_{it} \frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}} - z_{it} \frac{\zeta_{q\eta}}{\sqrt{T}}) - \rho(u_{it})]$ then

$$\begin{aligned}
P_2 &= P\left( \max_{1 \leq q \leq K_N} \frac{1}{NT} \left| \sum_i \sum_t [m_{it} - Em_{it}] \right| > \frac{\epsilon}{2} \right) \\
&\leq \sum_{q=1}^{K_N} P\left( \frac{1}{NT} \left| \sum_i \left[ \sum_t m_{it} - E \sum_t m_{it} \right] \right| > \frac{\epsilon}{2} \right).
\end{aligned}$$

For fixed $i$,

$$m_{it} \le \sup_{\zeta\in\Gamma}\left|\sum_k\left[\rho\left(u_{it}-\tilde{x}_{it}\frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}}-z_{it}\frac{\zeta_{q\eta}}{\sqrt{T}}\right)-\rho(u_{it})\right]\right|$$
$$\le \sup_{\zeta\in\Gamma}2\left|\sum_k\left[\tilde{x}_{it}\frac{\zeta_{q\tilde{\beta}}}{\sqrt{NT}}+z_{it}\frac{\zeta_{q\eta}}{\sqrt{T}}\right]\right|$$
$$\le \sup_{\zeta\in\Gamma}2\sum_k\left[\frac{1}{\sqrt{NT}}\|\tilde{x}_{it}\|\,\|\zeta_{q\tilde{\beta}}\|+\frac{1}{\sqrt{T}}\|z_{it}\|\,\|\zeta_{q\eta}\|\right]\le 2C_2,$$

and,

$$Var\left(\sum_{t=1}^T m_{it}\right)\le\sum_{t=1}^T Var(m_{it})+2\sum_{t_1<t_2}Cov(m_{it_1},m_{it_2})$$
$$\le\sum_{t=1}^T Var(m_{it})+\sum_{t_1<t_2}[Var(m_{it_1})+Var(m_{it_2})]$$
$$\le T\sum_{t=1}^T Var(m_{it}).$$

Note that,

$$\left|\frac{\tilde{x}_{it}}{\sqrt{NT}}\zeta_{q\tilde{\beta}}+\frac{z_{it}}{\sqrt{T}}\zeta_{q\eta}\right|^2\le C_3\|\zeta_{q\tilde{\beta}}\|+C_4\|\zeta_{q\eta}\|.$$

Using the fact that $\|\zeta_q\|<C$ for any $q$, and the independence across individuals in A1, we have

$$Var\sum_{i=1}^N\left(\sum_{t=1}^T m_{it}\right)\le T\sum_{i=1}^N\sum_{t=1}^T Ey_{it}^2\le NT^2C_5+o(1).$$

By the assumption of independence, we can bound $P_2$ using Bernstein's inequality:

$$P_2\le 2K_N\exp\left(-\frac{(\epsilon NT/2)^2}{\sum_i Var\sum_t m_{it}+\frac{TC_2}{3}\left(\frac{\epsilon NT}{2}\right)}\right)$$
$$\le 2\left(2\frac{\sqrt{p_N}}{q_{NT}}+1\right)^{p_N}\exp\left(-\frac{(\epsilon NT/2)^2}{NT^2C_5+o(1)+\frac{TC_2}{3}\left(\frac{\epsilon NT}{2}\right)}\right)$$
$$=2\exp\Bigg(O(N)\ \ln(24O(N^{1/2-a}T)C_1/\epsilon+1)$$
$$-\frac{(\epsilon NT/2)^2}{NT^2C_5+o(1)+\frac{TC_2}{3}\left(\frac{\epsilon NT}{2}\right)}\Bigg).$$

Hence, $P_2\to 0$ as $N,T\to\infty$ and $N^a/T\to 0$ and the lemma follows. □

Let $\theta(\tau)=(\theta(\tau_1)',\ldots,\theta(\tau_K)')'$. Now we prove Theorem 1.

**Proof.** The first part follows from Lemma 1. The second part is similar to Chernozhukov and Hansen (2006), we need to show that under conditions A1–A6, $\hat{\theta}(\tau)=\theta(\tau)+o_p(1)$. Let

$$\mathscr{P}=(\eta,\alpha,\beta,\gamma)\to\rho_\tau(y-Z\eta-y_{-1}\alpha-X\beta-W\gamma)$$

and note that $\mathscr{P}$ is continuous. Therefore, uniform convergence is given by Lemma 2, such that $\sup_{\theta\in\Theta}|M_n-M|=o_p(1)$, where $M_n\equiv\frac{1}{nT}\sum_{k=1}^K\sum_{i=1}^n\sum_{t=1}^T\rho(u_{it}-\tilde{x}_{it}\frac{\delta_{\tilde{\beta}}}{\sqrt{NT}}-z_{it}\frac{\delta_\eta}{\sqrt{T}})-\rho(u_{it})$, and $M=EM_n$. Therefore, denoting $\vartheta=(\eta,\beta,\gamma)$, we have that $\|\hat{\vartheta}(\alpha,\tau)-\vartheta(\alpha,\tau)\|\xrightarrow{p}0$ (*) which implies that $\|\,\|\hat{\gamma}(\alpha,\tau)\|-\|\gamma(\alpha,\tau)\|\,\|\xrightarrow{p}0$, which by a simple Argmax process over a compact set argument (Assumption A4 and Corollary 3.2.3 in van der Vaart and Wellner (1996)) implies that $\|\hat{\alpha}(\tau)-\alpha(\tau)\|\xrightarrow{p}0$, which by (*) implies that $\|\hat{\beta}(\tau)-\beta(\tau)\|\xrightarrow{p}0$ and $\|\hat{\gamma}(\hat{\alpha},\tau)-0\|\xrightarrow{p}0$. Therefore, $\|\hat{\theta}(\tau)-\theta(\tau)\|\xrightarrow{p}0$ and the theorem follows. □

Now we prove Theorem 2.

**Proof.** Consider the following model

$$y_{it}=\eta_i+\alpha y_{it-1}+\beta x_{it}+u_{it}.$$

The objective function is

$$\min_{\eta_i,\alpha,\beta,\gamma}\sum_{k=1}^K\sum_{i=1}^N\sum_{t=1}^T\upsilon_k\rho_\tau(y_{it}-\eta_i$$
$$-\alpha(\tau_k)y_{it-1}-\beta(\tau_k)x_{it}-\gamma(\tau_k)w_{it}).$$

Consider a collection of closed balls $B_n(\alpha(\tau_k))$, centered at $\alpha(\tau_k)$, with radius $\pi_n$, and $\pi_n\to 0$ slowly enough. Note that, for any $\alpha_n(\tau_k)\xrightarrow{p}\alpha(\tau_k)(\delta_\alpha\xrightarrow{p}0)$ we can write the objective function as

$$V_{NT}(\delta)=\sum_{k=1}^K\sum_{i=1}^N\sum_{t=1}^T\upsilon_k\rho_\tau(y_{it}-\xi_{it}(\tau_k)$$
$$-z_{it}\delta_\eta/\sqrt{T}-y_{it-1}\delta_\alpha/\sqrt{NT}$$
$$-x_{it}\delta_\beta/\sqrt{NT}-w_{it}\delta_\gamma/\sqrt{NT})$$
$$-\upsilon_k\rho_\tau(y_{it}-\xi_{it}(\tau_k))$$

where $\xi_{it}(\tau_k)=\eta_i+\alpha(\tau_k)y_{it-1}+\beta(\tau_k)x_{it}+\gamma(\tau_k)w_{it}$, and

$$\delta_n=\begin{pmatrix}\delta_\eta\\\delta_\alpha\\\delta_\beta\\\delta_\gamma\end{pmatrix}=\begin{pmatrix}\sqrt{T}(\hat{\eta}(\alpha_n)-\eta)\\\sqrt{NT}(\alpha_n(\tau_k)-\alpha(\tau_k))\\\sqrt{NT}(\hat{\beta}(\alpha_n,\tau_k)-\beta(\tau_k))\\\sqrt{NT}(\hat{\gamma}(\alpha_n,\tau_k)-0)\end{pmatrix}.$$

Note that, for fixed $(\delta_\alpha,\delta_\beta,\delta_\gamma)$, we can consider the behavior of $\delta_\eta$. Let $\psi(u)\equiv(\tau-I(u<0))$ and for each $i$

$$g_i(\delta_\eta,\delta_\alpha,\delta_\beta,\delta_\gamma)=-\frac{1}{\sqrt{T}}\sum_{k=1}^K\sum_{t=1}^T\upsilon_k\psi_\tau$$
$$\times\Bigg(y_{it}-\xi_{it}-\frac{\delta_\eta}{\sqrt{T}}-\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1}$$
$$-\frac{\delta_\beta}{\sqrt{NT}}x_{it}-\frac{\delta_\gamma}{\sqrt{NT}}w_{it}\Bigg).$$

For, fixed $\delta_\beta,\delta_\gamma$, $\sup_{\tau\in t}\|\alpha_n(\tau)-\alpha(\tau)\|\xrightarrow{p}0$, and $K>0$

$$\sup_{\|\delta\|<K}\|g_i(\delta_{\eta_i},\delta_\alpha,\delta_\beta,\delta_\gamma)-g_i(0,0,0,0)$$
$$-E[g_i(\delta_\eta,\delta_\alpha,\delta_\beta,\delta_\gamma)-g_i(0,0,0,0)]\|=o_p(1).$$

Expanding we have

$$E[g_i(\delta_\eta,\delta_\alpha,\delta_\beta,\delta_\gamma)-g_i(0,0,0,0)]$$
$$=E\Bigg(-\frac{1}{\sqrt{T}}\sum_{k=1}^K\sum_{t=1}^T\upsilon_k\psi_\tau$$
$$\times\left(y_{it}-\xi_{it}-\frac{\delta_\eta}{\sqrt{T}}-\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1}-\frac{\delta_\beta}{\sqrt{NT}}x_{it}-\frac{\delta_\gamma}{\sqrt{NT}}w_{it}\right)$$
$$+\frac{1}{\sqrt{T}}\sum_{k=1}^K\sum_{t=1}^T\upsilon_k\psi_\tau(y_{it}-\xi_{it})\Bigg)$$
$$=-\frac{1}{\sqrt{T}}\sum_{k=1}^K\sum_{t=1}^T\upsilon_k\left[\tau_k-F\left(\xi_{it}+\frac{\delta_\eta}{\sqrt{T}}+\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1}\right.\right.$$

$$+ \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it}\Big)\Big]$$

$$= \frac{1}{\sqrt{T}}\sum_{k=1}^{K}\sum_{t=1}^{T}\upsilon_k f_{it}(\xi_{it}(\tau_k))\left[\frac{\delta_\eta}{\sqrt{T}} + \frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} + \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it}\right] + R_{it}.$$

Optimality of $\hat{\delta}_{\eta_i}$ implies that $g_i(\delta_\eta, \delta_\alpha, \delta_\beta, \delta_\gamma) = o(T^{-1})$, and thus

$$\hat{\delta}_\eta = -\bar{f}_i^{-1}\Bigg[\frac{1}{\sqrt{T}}\sum_{k=1}^{K}\sum_{t=1}^{T}\upsilon_k f_{it}(\xi_{it}(\tau_k)) \times \left[\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} + \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it}\right] - \frac{1}{\sqrt{T}}\sum_{k=1}^{K}\sum_{t=1}^{T}\upsilon_k\psi_\tau(y_{it} - \xi_{it}(\tau_k))\Bigg] + R_{it},$$

where $\bar{f}_{ik} = T^{-1}\sum_k\sum_t \upsilon_k f_{it}$ and $R_{it}$ is the remainder term for each $i$. Substituting $\hat{\delta}_\eta$'s, we denote

$$G(\delta_\alpha, \delta_\beta, \delta_\gamma) = -\frac{1}{\sqrt{NT}}\sum_{k}^{K}\sum_{i=1}^{N}\sum_{t=1}^{T}\upsilon_k X_{it}\psi_\tau \times \left(y_{it} - \xi_{it} - \frac{\hat{\delta}_\eta}{\sqrt{T}} - \frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} - \frac{\delta_\beta}{\sqrt{NT}}x_{it} - \frac{\delta_\gamma}{\sqrt{NT}}w_{it}\right),$$

where $X_{it} = (x'_{it}, w'_{it})'$, and $\delta_\alpha(\alpha_n) = \sqrt{NT}(\alpha_n(\tau_k) - \alpha(\tau_k))$, $\delta_\beta(\alpha_n) = \sqrt{NT}(\hat{\beta}(\alpha_n, \tau_k) - \beta)$, and $\delta_\gamma(\alpha_n) = \sqrt{NT}(\hat{\gamma}(\alpha_n, \tau_k) - 0)$. As in Koenker (2004)

$$\sup_{\|\delta\|<K}\|G(\delta_\alpha, \delta_\beta, \delta_\gamma) - G(0,0,0) - E[G(\delta_\alpha, \delta_\beta, \delta_\gamma) - G(0,0,0)]\| = o_p(1),$$

and at the minimizer, $G(\hat{\delta}_\alpha, \hat{\delta}_\beta, \hat{\delta}_\gamma) = o((NT)^{-1})$. Expanding, as above,

$$\begin{aligned}
&E[G(\delta_\alpha, \delta_\beta, \delta_\gamma) - G(0,0,0)] \\
&= \frac{1}{\sqrt{NT}}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} \times \left(\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} + \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it} + \frac{\hat{\delta}_\eta}{\sqrt{T}}\right) + o_p(1) \\
&= \frac{1}{\sqrt{NT}}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} \times \Bigg[\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} + \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it} \\
&\quad - \frac{1}{\sqrt{T}}\bar{f}_i^{-1}\Bigg[T^{-1/2}\sum_k\sum_t \upsilon_k f_{it}(\xi_{it}(\tau_k)) \times \left[\frac{\delta_\alpha}{\sqrt{NT}}y_{it-1} + \frac{\delta_\beta}{\sqrt{NT}}x_{it} + \frac{\delta_\gamma}{\sqrt{NT}}w_{it}\right] \\
&\quad - T^{-1/2}\sum_k\sum_{t=1}^{T}\upsilon_k\psi_\tau(y_{it} - \xi_{it}(\tau_k))\Bigg] + R_{it}\Bigg] + o_p(1) \\
&= \frac{1}{NT}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} \times \left(y_{it-1} - \bar{f}_i^{-1}T^{-1}\sum_k\sum_t \upsilon_k f_{it} y_{it-1}\right)\delta_\alpha \\
&\quad + \frac{1}{NT}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} \times \left(x_{it} - \bar{f}_i^{-1}T^{-1}\sum_k\sum_t \upsilon_k f_{it} x_{it}\right)\delta_\beta \\
&\quad + \frac{1}{NT}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} \times \left(w_{it} - \bar{f}_i^{-1}T^{-1}\sum_k\sum_t \upsilon_k f_{it} w_{it}\right)\delta_\gamma \\
&\quad - \frac{1}{\sqrt{NT}}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it}\bar{f}_i^{-1}T^{-1/2} \times \sum_k\sum_t \upsilon_k\psi(y_{it} - \xi_{it}) \\
&\quad - \frac{1}{\sqrt{NT}}\sum_k\sum_i\sum_t \upsilon_k X_{it} f_{it} R_{it}/\sqrt{T} + o_p(1),
\end{aligned}$$

where the order of the final term is controlled by the bound on the derivative of the conditional density. It is important to note that $G(\hat{\delta}_\alpha, \hat{\delta}_\beta, \hat{\delta}_\gamma) = 0$ and then $E[G(\delta_\alpha, \delta_\beta, \delta_\gamma) - G(0,0,0)] = G(0,0,0)$, $\Phi = \mathrm{diag}(f_{it}(\xi_{it}(\tau_k)))$, and let $\Psi_\tau$ be a $NT$-vector $(\psi_\tau(y_{it} - \xi_{it}(\tau)))$. Define $\delta_\vartheta = (\delta_\beta, \delta_\gamma)$, then omitting the subscript $k$ we have

$$\begin{aligned}
&\upsilon(X'M_Z\Phi M_Z Y_{-1}/NT)\delta_\alpha + \upsilon(X'M_Z\Phi M_Z X/NT)\delta_\vartheta \\
&\quad - \upsilon X'P_Z\Psi/\sqrt{NT} = -\upsilon X'\Psi/\sqrt{NT} + R_{nT} \\
&J_\vartheta\delta_\vartheta = [-(\upsilon X'M_Z\Psi/\sqrt{NT} - R_{nT}) - J_\alpha\delta_\alpha] \\
&\hat{\delta}_\vartheta = J_\vartheta^{-1}[-(\upsilon X'M_Z\Psi/\sqrt{NT} - R_{nT}) - J_\alpha\delta_\alpha]
\end{aligned}$$

where $M_Z = I - P_Z$, $P_Z = Z(Z'\Phi Z)^{-1}Z'\Phi$, $J_\vartheta$ and $J_\alpha$ are as in assumption A3, and

$$R_{nT} = \frac{1}{\sqrt{NT}}\sum_{k=1}^{K}\sum_{i=1}^{N}\sum_{t=1}^{T}\upsilon_k X_{it} f_{it} R_{it}/\sqrt{T} + o_p(1).$$

Let $[\bar{J}'_\beta, \bar{J}'_\gamma]'$ be the conformable partition of $J_\vartheta^{-1}$, then

$$\hat{\delta}_\gamma = \bar{J}_\gamma[-(\upsilon X'M_Z\Psi/\sqrt{NT} - R_{nT}) - J_\alpha\delta_\alpha] \qquad (16)$$

$$\hat{\delta}_\beta = \bar{J}_\beta[-(\upsilon X'M_Z\Psi/\sqrt{NT} - R_{nT}) - J_\alpha\delta_\alpha]. \qquad (17)$$

The remainder term $R_{nT}$ has a dominant component that comes from the Bahadur representation of the $\eta$'s. By A1 and A5, we have for a generic constant $C_1$

$$R_{nT} = T^{-1/4}\frac{C_1 K}{\sqrt{N}}\sum_{i=1}^{N}R_{i0} + o_p(1).$$

The analysis of Knight (2001) shows that the summands converge in distribution, that is as $T \to \infty$ the remainder term $T^{1/4}R_{it} \xrightarrow{d} R_{i0}$, where $R_{i0}$ are functionals of Brownian motions with finite second moment. Therefore, independence of $y_{it}$, and condition A6 ensure that the contribution of the remainder is negligible. Thus (16) and (17) simplify to

$$\hat{\delta}_\gamma = \bar{J}_\gamma[-(\upsilon X'M_Z\Psi/\sqrt{NT}) - J_\alpha\delta_\alpha],$$

$$\hat{\delta}_\beta = \bar{J}_\beta[-(\upsilon X'M_Z\Psi/\sqrt{NT}) - J_\alpha\delta_\alpha].$$

By consistency, $wp \to 1$

$$\hat{\delta}_\alpha = \min_{\delta_\alpha\in B_n(\alpha(\tau))}\hat{\delta}_\gamma(\delta_\alpha)'A\hat{\delta}_\gamma(\delta_\alpha),$$

assuming that $\hat{\delta}_{\gamma}' A \hat{\delta}_{\gamma}$ is continuous in $\delta_\alpha$, and from the first order condition

$$\hat{\delta}_{\alpha} = -[J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma J_\alpha]^{-1} [J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma (\upsilon X' M_Z \Psi / \sqrt{NT})].$$

Substituting $\hat{\delta}_\alpha$ back in $\delta_\beta$ we obtain

$$\begin{aligned}\hat{\delta}_{\beta} &= -\bar{J}_\beta [(X' M_Z \Psi / \sqrt{NT}) \\ &\quad - J_\alpha [J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma J_\alpha]^{-1} J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma (\upsilon X' M_Z \Psi / \sqrt{NT})] \\ &= -\bar{J}_\beta [(I - J_\alpha [J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma J_\alpha]^{-1} J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma) \\ &\quad \times (\upsilon X' M_Z \Psi / \sqrt{NT})].\end{aligned}$$

It is also important to analyze $\hat{\delta}_\gamma$. Thus, replacing $\hat{\delta}_\alpha$ in $\delta_\gamma$

$$\begin{aligned}\hat{\delta}_{\gamma} &= -\bar{J}_\gamma [(X' M_Z \Psi / \sqrt{NT}) - J_\alpha [J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma J_\alpha]^{-1} \\ &\quad \times J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma (\upsilon X' M_Z \Psi / \sqrt{NT})] \\ &= -\bar{J}_\gamma [(I - J_\alpha [J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma J_\alpha]^{-1} J_\alpha' \bar{J}_\gamma A \bar{J}_\gamma) \\ &\quad \times (\upsilon X' M_Z \Psi / \sqrt{NT})].\end{aligned}$$

By condition A3, using the fact that $\bar{J}_\gamma J_\alpha$ is invertible

$$\hat{\delta}_{\gamma} = 0 + O_p(1) + o_p(1).$$

Let $\Psi_k = \mathrm{diag}(\psi_{\tau_k}(y_{it} - \xi_{it}(\tau_k)))$. Notice that $\Psi_k 1_{NT} 1_{NT}' \Psi_l = (\tau_k \wedge \tau_l - \tau_k \tau_l) I_{NT}$, and that conditions A1–A6 imply a central limit theorem. Thus, neglecting the remainder term, and using the definition of $\delta_\alpha$ and $\delta_\beta$ we have

$$\sqrt{NT}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Omega), \quad \Omega = (K', L')' S (K', L')$$

where $S = (\tau \wedge \tau' - \tau \tau') E(VV')$, $V = X' M_Z$, $K = (J_\alpha' H J_\alpha)^{-1} J_\alpha H$, $H = \bar{J}_\gamma' A[\alpha(\tau)] \bar{J}_\gamma$, $L = \bar{J}_\beta M$, $M = I - J_\alpha K$. $\square$

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